Intelligent Detection of Sports Fields Based on Very High-Resolution Stereo Satellite Imagery

Dashuai Shang, Yongsheng Zhang, Chenguang Dai, and Lei Zhang

***Abstract*—Sports fields are primary locations for people to engage in sports activities and cultural exchanges, possessing significant social and economic value and being an important object for remote-sensing mapping. To extract sports field data efficiently and accurately, the first multicategory, fine-grained sports field dataset, SF_VHR_SSI, was constructed based on various measured very-high-resolution stereo satellite data. It includes comprehensive metadata information and diverse sample data, capable of providing data support for various types of object detection tasks. To ascertain the scientific validity and rationality of the dataset, extensive benchmarking experiments were conducted utilizing 15 established algorithms encompassing 54 distinct models. Based on these experiments, the complexity and challenges inherent in the SF_VHR_SSI dataset were analyzed, thereby providing insights that inform the design of subsequent algorithms. To improve the accuracy of sports field detection further, partial convolution (PConv) was introduced, a squeeze-and-excitation (SE) mechanism was incorporated, and a multidetection box fusion strategy was employed. The FasterC-SPELAN4 feature extraction module, SEAttention attention module, and Merge-NMS postprocessing module were designed. The novel SF_GELAN network was constructed based on the generalized efficient layer aggregation network (GELAN) architecture, which is both efficient and accurate. Furthermore, by utilizing model scaling techniques, three versions of the network-large, medium, and small-were developed to cater to diverse application scenarios. A comparison of the experimental results demonstrates that all three versions of the SF_GELAN network achieved the state-of-the-art performance on the SF_VHR_SSI dataset, providing robust algorithmic support for high-precision sports field detection globally.**

***Index Terms*—Multidetection box fusion, object detection, partial convolution (PConv), sports field dataset, squeeze-and-excitation (SE) mechanism, very-high-resolution satellite stereo imagery.**

I. Introduction

With the advancement of the global economy, enhancement of public health awareness, and widespread participation in sports activities, a significant annual increase in the quantity and expanse of sports fields has been observed [1], [2], [3]. They have become an integral part of urban public infrastructure, and the application of remote-sensing and artificial-intelligence technologies for their surveillance and management is gaining interest. Recently, with the maturation of technologies, such as sensors, satellites, and aerial vehicles, the volume of remote-sensing data has grown exponentially and the spatial resolution of imagery has been enhanced from medium resolution (4–70 m) to high resolution (1–4 m) and now to very high resolution (>1 m) [4]. The current very-high-resolution stereo remote-sensing satellites include the WorldView2 (0.46 m) [5] and WorldView3 (0.31 m) [6] owned by the USA, Pleiades (0.50 m) [7] and Pleiades Neo (0.30 m) [8] owned by France, and Gaofen-7 (0.65 m) [9] and Gaofen-14 (0.60 m) [10] owned by China. The submeter very-high-resolution stereo imagery data acquired by these satellites contain rich texture information and possess high-precision geospatial information, making high-precision detection of targets, such as sports fields possible. Currently, approximately 95% of sports fields are built by enterprises, institutions, and government agencies [3], serving as primary locations for public engagement in sports activities and cultural exchanges and having significant social and economic value. In the context of remote-sensing imagery, sports fields are considered typical geospatial features, characterized by their widespread distribution, flat terrain, and distinct linear characteristics, making them an important subject for mapping and an ideal carrier for control information transmission, which confers substantial value for remote-sensing applications. Consequently, research on intelligent extraction technology for sports fields based on very-high-resolution stereo imagery not only helps in the construction and management of smart cities but also provides technical support for high-precision applications, such as intelligent remote-sensing mapping [11] and the construction of a global intelligent landmark control network.

Intelligent extraction of sports fields from very-high-resolution stereo imagery utilizes deep-learning-based object detection techniques to extract sports field image patches from high-resolution stereo images. This process primarily involves the construction of sports field datasets and the design of intelligent extraction algorithms. Sports fields are considered typical geospatial feature datasets in remote-sensing imagery, garnering widespread attention in the field of object detection. Notable remote-sensing image object detection datasets that include sports venues are NWPU VHR-10 [12], RSOD [13], DOTA [14], DIOR [15], and FAIRAM [16].

The NWPU VHR-10 dataset encompasses ten categories with 800 images, primarily sourced from Google Images,

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Dashuai Shang is with the Geospatial Information College, Information Engineering University, Zhengzhou 450001, China, also with the 61363 Troops, Chinese People's Liberation Army, Xi'an 710054, China, and also with the Key Laboratory of Smart Earth, Xi'an 710054, China (e-mail: dad121jm@163.com).

Yongsheng Zhang, Chenguang Dai, and Lei Zhang are with the Geospatial Information College, Information Engineering University, Zhengzhou 450001, China.

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with spatial resolutions ranging from 0.5 to 2 m. Eighty-one images contain sports field objects, with varying image sizes of approximately 1000 × 700 pixels, totaling 390 baseball field instances, 524 tennis court instances, 159 basketball court instances, and 163 ground track field instances. The RSOD dataset encompasses four classes with 976 images, primarily sourced from Google Earth and Tianditu imagery. Of the images, 189 contain sports field objects of varying sizes, approximately 1000 × 900 pixels, comprising 191 ground track field instances. The DOTA dataset, primarily used for object detection in aerial imagery, has expanded into version 2.0 to include data from Google Earth and Gaofen-2 imagery. The DIOR dataset comprises 20 classes with 23 463 images derived from Google Earth, with a uniform size of 800 × 800 pixels and a spatial resolution ranging from 0.5 to 30 m. Among the images, 5144 include sports field objects, with 3568 instances of baseball fields, 2034 of basketball courts, 3302 of track fields, and 3817 of tennis courts. The FAIRAM dataset includes five major categories, 37 subcategories, and more than 15 000 images sourced from Google Earth and Gaofen orthoimages, with a spatial resolution between 0.3 and 0.8 m. Approximately 1586 images contain sports field objects of varying sizes, encompassing four subcategories: tennis courts, football fields, basketball courts, and baseball fields.

Although these datasets include sports fields, they cannot be used in actual engineering scenarios. First, the image sizes within these datasets are not uniform [12], [13], [16], necessitating image scaling during algorithm training, leading to potential information loss or waste of computational resources and deviating from actual engineering applications, where the remote-sensing imagery is typically cropped into fixed-sized image patches for object detection. Second, the images in these datasets are primarily sourced from Google Earth [12], [13], [14], [15], [16], which predominantly offers color orthoimages; however, high-precision applications, such as remote-sensing mapping and control baseline construction, are based on high-resolution raw panchromatic imagery and not color orthoimages. Furthermore, the sports field categories in these datasets are not comprehensive, with at most only four categories, whereas the diversity of sports field types is greater in real-life scenarios. Finally, because Google Earth is the origin of these datasets, metadata, such as spatial resolution and imaging time, are incomplete and unknown, failing to provide support for subsequent algorithmic analysis and improvement; for instance, in the highly regarded DOTA dataset, only 30% of the images have spatial resolution information [14]. To address these issues, in this study, a sports field dataset (SF_VHR_SSI) was created based on 1B-level imagery products from actual very-high-resolution satellite stereo imagery, which is characterized by uniform size, comprehensive types, and complete metadata (detailed information can be found in Section II).

Deep-learning-based object detection techniques autonomously learn to extract features, capturing more abstract and higher dimensional characteristics of targets with greater accuracy, making them the preferred choice for intelligent extraction algorithms. Depending on their processes, these methods can be categorized into two-stage object detection methods based on candidate regions and one-stage object detection methods based on regression [4], [17].

Two-stage object detection algorithms select candidate regions from the image and then perform regression localization and classification for potential targets within the regions, dividing the overall detection process into two distinct phases. Examples include those in the region-based convolutional neural network (R-CNN) series—faster R-CNN [18], mask R-CNN [19], and cascade R-CNN [20]. The R-CNN series represents the earliest deep-learning-based object detection algorithms, evolving from R-CNN [21] to fast R-CNN [22] and then to faster R-CNN [18]. Faster R-CNN significantly improved detection speed by employing a region proposal network, becoming the first deep-learning-based object detection algorithm to achieve end-to-end training. However, it faces issues, such as low detection accuracy and poor performance in detecting small targets. Subsequently, measures, such as introducing feature pyramid networks (FPNs) [23], increasing the model depth of the network [24] (utilizing ResNet101 as the backbone network), improving the accuracy of feature map alignment [19] (Mask R-CNN), refining training methods [20] (cascade R-CNN), and addressing data imbalance issues [25] (Libra R-CNN), were implemented to enhance detection performance. The incorporation of FPN [23] is nearly a standard feature in later algorithm designs.

One-stage object detection algorithms do not separately select candidate regions, omitting the candidate region generation step and integrating feature extraction, object classification, and location regression into a single operational phase. Examples include the you only look once (YOLO) series (YOLOv1–YOLOv11), single shot multibox detector series, and RetinaNet series. The YOLO series is the earliest and most maturely developed one-stage object detection algorithm. YOLOv1 [26] featured high detection efficiency, achieving real-time object detection, albeit with relatively low accuracy. Subsequent versions have primarily focused on improving detection accuracy using anchor box mechanisms (YOLOv2 [27]), increasing model depth (YOLOv3 [28]), refining data augmentation (YOLOv4 [29]), enhancing feature fusion (YOLOv4 [29] and YOLOv5 [30]), optimizing loss functions (YOLOv6 [31] and YOLOv8 [32]), adding network branches (YOLOv9 [33]), improving detection heads (YOLOv6 [31] and YOLOv7 [34]), and optimizing post-processing (YOLOv10 [35]). These efforts aim to balance detection accuracy with detection efficiency. RetinaNet [36] addresses the extreme imbalance between positive and negative samples during the training process of one-stage object detection algorithms by redefining the standard cross-entropy loss and designing a new loss function—focal loss [36], which has achieved commendable detection results. Focal loss is almost a standard component in one-stage algorithms.

Several of these algorithms incorporate anchor boxes, which require the design of parameters, such as scale, quantity, and aspect ratio, based on prior knowledge. This approach suffers from poor generalization, difficulty in detecting objects of unusual sizes, and imbalance between positive and negative samples. Consequently, researchers have introduced anchor-

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free detection, leading to the development of algorithms, such as CornerNet [37], CenterNet [38], and fully convolutional one-stage object detection (FCOS) [39], that do not rely on anchor boxes. CornerNet [37] reformulates the object detection problem into a keypoint prediction problem, locating bounding boxes by predicting the positions of the top-left and bottom-right corners. Building upon CornerNet [37], CenterNet [38] includes the prediction of the center point, enabling the algorithm to consider features within the bounding box and enhancing detection accuracy. FCOS [39] adopts a pixelwise prediction approach for bounding boxes, achieving object localization and classification by predicting the distances from a pixel to its corresponding object boundaries and the probabilities of the object belonging to different categories. Compared to anchor-based detection algorithms, the anchor-free detection algorithms exhibit stronger generalization capabilities and perform better in scenarios involving small and multiscale targets but offer lower accuracy in general object detection tasks [39].

The aforementioned classical object detection algorithms are general purpose. Later developments have focused on specific application scenarios, such as transportation, agriculture, remote sensing, and industry. Improvements have been proposed based on these classical algorithms to enhance accuracy and efficiency in specific datasets and application contexts. The specific improvement strategies can be categorized into several areas.

1) *Enhancing Data Augmentation Strategies:* For instance, leveraging diffusion models to increase high-level semantic diversity in data [40] and employing denoising [41] and defogging techniques [42] to improve data quality.
2) *Optimizing the Backbone Network:* Introducing large kernel convolutions (LSKNet [43]) and multiscale convolutions (PKINet [44]) to capture contextual information and enhance the ability of the network to extract multiscale features and incorporating hybrid attention [45] and self-attention (YOLOv10 [35]) modules to improve the generalization capabilities of the network.
3) *Adding Auxiliary Network Branches:* For example, introducing an auxiliary reversible branch to generate reliable gradient information to support backpropagation in the main branch (YOLOv9 [33]) and incorporating a super resolution branch to enhance the detection capacity for small targets (SuperYOLO [46]).
4) *Strengthening Feature Fusion:* Employing an aggregate–distribute mechanism for global multilevel feature fusion and reducing information loss during the feature fusion process (Gold-YOLO [47]) and enhancing feature fusion through dense information exchange between features of different scales (GiraffeDet [48]).
5) *Optimizing Loss Functions:* Adding penalty terms for aspect ratio (CIoU [49]), orientation (SIoU [50]), and shape (Shape-IoU [51]) to improve the accuracy of bounding box regression and incorporating weight information (Wise-IoU [52] and Focaler-IoU [53]) to enhance the focus of the network on specific samples.
6) *Improving Postprocessing Techniques:* Introducing adaptive mechanisms [54] that automatically adjust nonmaximum suppression (NMS) thresholds based on the density of targets and incorporating weight information [55] to achieve weighted fusion of multiple high-confidence bounding boxes during NMS processing, thereby enhancing the regression accuracy of bounding boxes.

These refinements are essential for tailoring object detection algorithms to meet the demands of specific applications, achieving high accuracy and efficiency in real-world scenarios.

Based on the SF_VHR_SSI dataset, benchmark experiments were implemented for 15 classic algorithms, including the R-CNN, YOLO, RetinaNet, and FCOS series. Building on this foundation and in conjunction with the characteristics of very-high-resolution remote-sensing satellite imagery, the SF_GELAN network was developed based on the generalized efficient layer aggregation network (GELAN) architecture [33], which balances efficiency and accuracy. This network integrates multiple networks and technologies, including FasterNet [56], SENetV2 [57], and Merge-NMS [18], [29], [30]. To cater to different application scenarios, SF_GELAN is designed in large, medium, and small versions. Therefore, the main contributions of this study are as follows.

1) A large-scale dataset, SF_VHR_SSI, was constructed based on ultrahigh-resolution raw stereo satellite imagery. It contains comprehensive metadata and diverse sample data, closely aligned with actual engineering applications. It categorizes sports fields into six classes, making it a fine-grained dataset [16], [58] that can support various types of object detection tasks, including fine-grained target detection and satellite imagery object detection.
2) To present the characteristics of SF_VHR_SSI objectively, extensive benchmark experiments were conducted on a general object detection platform, evaluating the performance of 54 network models on SF_VHR_SSI, thus validating its rationality and scientific nature. By combining the experimental results, the complexity and challenges were analyzed, providing guidance for designing subsequent satellite imagery-based object detection algorithms.
3) Because of the characteristics of SF_VHR_SSI and the requirements of actual engineering applications, the SF_GELAN network is proposed based on the lightweight GELAN architecture. This network incorporates partial convolution (PConv) [56] to design feature extraction modules, enhancing network efficiency while improving detection accuracy. Furthermore, a squeeze-and-excitation (SE) mechanism [57] and a multidetection box fusion mechanism are added to refine accuracy, achieving synchronized enhancement and effective balance of efficiency and precision and serving as a benchmark for subsequent algorithm improvements.

II. Details of the SF_VHR_SSI Dataset

A. Image Collection and Preprocessing

In satellite imagery, the sensors used for image acquisition, the spatial resolution of the imagery, and the product level of the image data are the primary factors contributing

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TABLE I
BASIC PARAMETERS OF VERY-HIGH-RESOLUTION STEREO SATELLITES AND THEIR IMAGERY

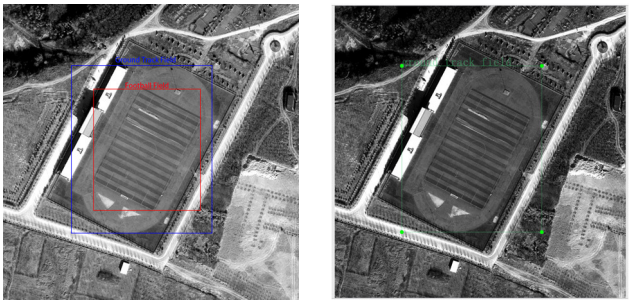
Satellite	WorldView2	WorldView3	Pleiades	BJ03N	GF07	GF14
Orbital Altitude	770 km	617 km	694 km	620 km	500 km	500 km
Image Size	37000 × 33000	35000 × 32000	50000 × 40000	28000 × 35000	33000 × 31000	69000 × 69000
Resolution	0.46 m	0.31 m	0.50 m	0.30 m	0.65 m	0.60 m
Swath Width	16 km	13 km	20 km	14 km	20 km	40 km
Band	450–800 nm	450–800 nm	470–830 nm	450–800 nm	450–900 nm	450–900 nm
Sample Count	252	146	344	33	828	2250

to dataset bias [59], [60]. The product level is a crucial determinant of the application direction of the dataset. High-resolution imagery directly obtained from satellites is typically panchromatic; after satellite operators receive the image data, it undergoes a series of automated processes, including decoding, scene segmentation, radiometric correction, and sensor correction, to form a 1B-level product [61], [62]. Such products are provided to surveying and mapping production departments and research institutions for the creation of high-precision geographic information products. The SF_VHR_SSI sample images are derived from 1B-level imagery products of high-resolution satellite stereo imagery and can be used in high-precision remote-sensing applications, such as geographic information product creation. To eliminate the influence of spatial resolution and sensors, imagery data were collected from six very-high-resolution stereo satellites—WorldView2, WorldView3, Pleiades, BJ03N (Pleiades NEO), GF07 (GaoFen-7), and GF14 (GaoFen-14)—to create the dataset. The basic parameters and image information of these satellites are listed in Table I.

The high-resolution stereo satellite imagery products consist of 16-bit panchromatic TIFF format images, with image sizes exceeding 20 000 pixels. Limited by current hardware capabilities, the existing algorithms cannot process samples with tens of thousands of pixels in a single input, necessitating image cropping. Considering the sample size requirements of the algorithms (typically 512 or 640 pixels) and the needs of future algorithm development, the SF_VHR_SSI dataset divides the original images into 1000- × 1000-pixel image patches for subsequent data processing. Furthermore, considering that mainstream algorithms currently only support 8-bit data, converting the 16-bit images to 8-bit images is necessary. The conversion process must not be a simple truncation or proportional compression because would result in significant loss of texture and radiance information.

B. Category Design and Definition

The proposed dataset is primarily aimed at the intelligent extraction of sports fields. It categorizes sports fields into six classes based on their frequency of occurrence and value in practical applications: basketball courts, ground track fields, football fields, tennis courts, hockey fields, and baseball diamonds. The basketball courts are standard basketball courts and exclude half-court and small courts (children's basketball courts). Football fields encompass futsal fields, seven-a-side football fields, standard football fields, and rugby fields. Ground track fields are primarily sports areas equipped with a circular track. In the case of multifunctional sports fields, the category with the larger area is taken as the class of that sports field. For instance, as shown in Fig. 1, when a football field overlaps with a ground track field, and the area of the ground track field is larger because it encloses the football field; the area is, thus, marked as a ground track field and the football field is not considered.



(a) (b)

Fig. 1. Schematics of multifunctional sports field annotation (a) raw image and (b) accurate annotation.

C. Image Annotation and Dataset Splits

This dataset employs widely applicable horizontal bounding boxes for annotation. The annotation process combines automatic labeling based on manual labeling with subsequent manual refinement. The steps are as follows.

First, a subset of samples was manually annotated using the LabelImg [63] tool to create an initial dataset, which includes 1181 image samples. To ensure the representativeness of the initial dataset, the annotated sample set must cover six different types of sports fields and six different image sources.

Second, the training environment was constructed (Table II) and training parameters were set (weights: yolov7.pt, epochs: 300, and batch size: 16, with all other parameters using the default values of the algorithm). The YOLOv7 [34] model was trained using the initial dataset for subsequent automatic annotation.

Subsequently, the well-trained YOLOv7 model was utilized for inference to achieve automatic annotation, and the annotation results were output as TXT files in the YOLO dataset format. During the inference process, the confidence threshold

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TABLE II
SPECIFICS OF THE TRAINING ENVIRONMENT

Environment	Components	Version/Model
Software	Platform	win32
	Python	3.8.19
	PyTorch	2.2.2
	CUDA	12.1
Hardware	GPU	NVIDIA GeForce RTX 4090 Laptop GPU
	CPU	AMD Ryzen 9 7945HX3D 2.30 GHz
	Memory	16G Dedicated GPU Memory + 16G Shared Memory

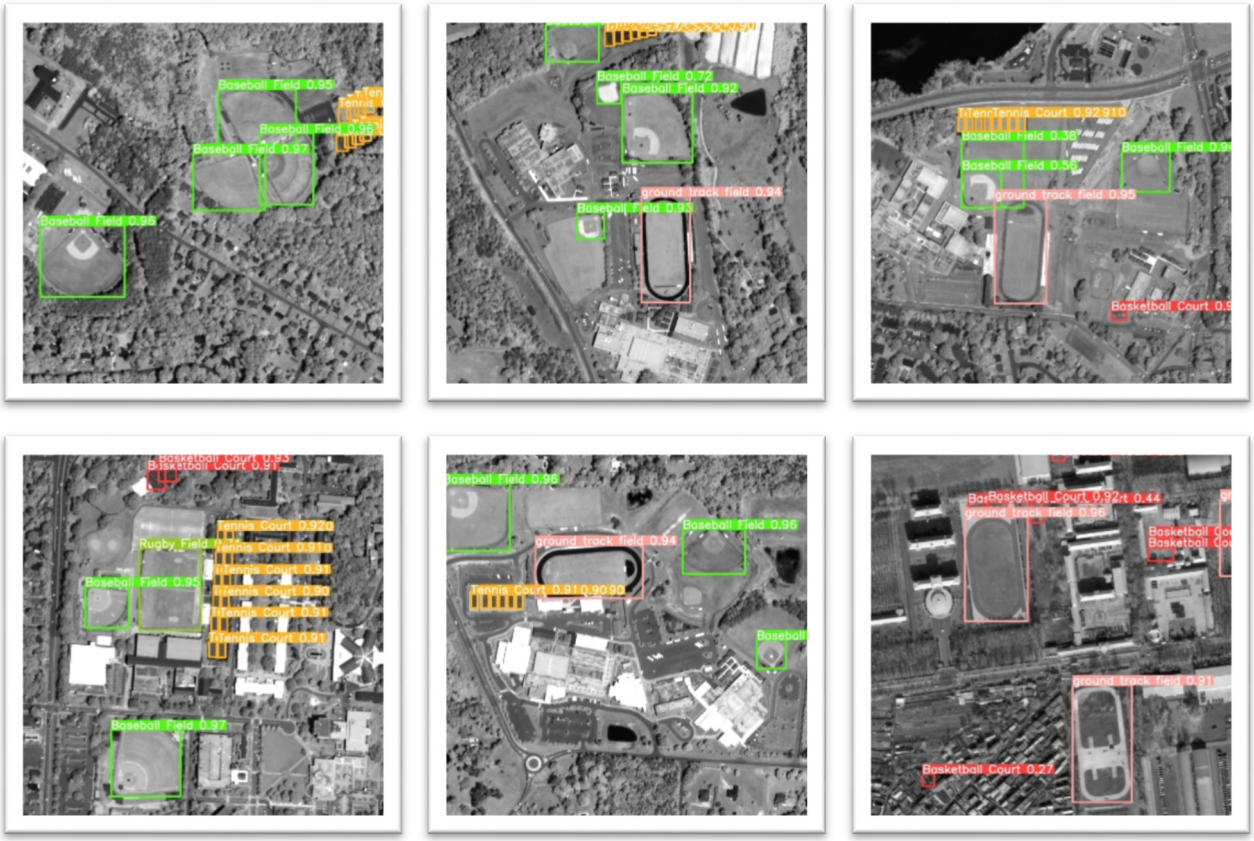


Fig. 2. Schematic of automatic sports field annotation results.

can be appropriately adjusted to enhance the ability of the model to detect sports fields. The automatic annotation results are shown in Fig. 2.

Finally, the automatic annotation results were read using the LabelImg tool and engineering technicians with extensive interpretation experience reviewed and corrected the automatic annotation results: first, to remove incorrectly identified sports fields, such as squares and farmlands; second, to add sports fields that were missed because of obstructions, such as buildings and vegetation; and third, to correct the positioning of inaccurately located sports fields.

Ultimately, 3853 sports field samples, encompassing 14 072 instances of various sports fields, were extracted from more than 100 high-resolution stereo image pairs covering multiple countries, including China, USA, South Africa, India, and Argentina. In the field of satellite Earth observation, the size of the images is extremely large (Table II); thus, the number of images does not reflect the scale of the dataset [14]. Therefore, comparing the scale between natural image and satellite image datasets using pixel area would be more appropriate.

To ensure the practicality of the dataset and enhance the generalization ability of the trained models, the division of the dataset must adhere to principles of randomness, independence, and representativeness. Therefore, the division process must consider factors, such as geography, image sources, sample quantity, number of instances, and class distribution. The division process is as follows. First, the dataset was divided into multiple subsets based on geography and image

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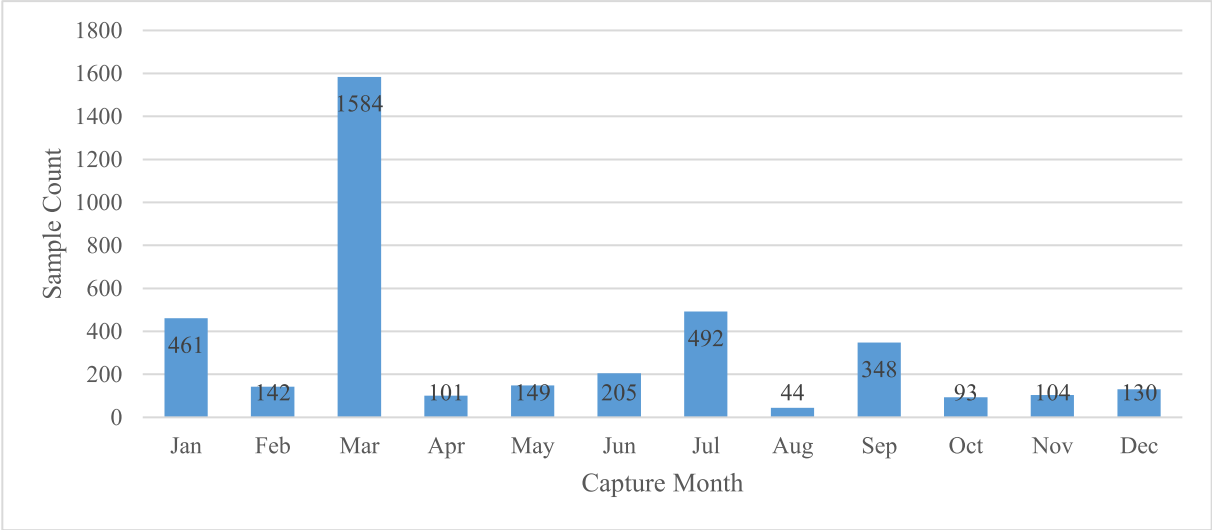


Fig. 3. Statistical chart of sample quantities obtained in different months of image capture.

sources. Second, within each subset, the samples were divided in a 6:2:2 ratio according to the sample quantity (multiple division outcomes were observed), and random numbers were used to generate 10 000 division outcomes (the exact number can be varied based on sample quantity). Subsequently, among the division outcomes, the one with the instance count ratio closest to 6:2:2 was selected as the division result. Finally, all subset division results were merged to construct the sports field dataset, which includes 2293 training samples, 780 validation samples, and 780 test samples. To maximize the utilization of sample data for training, a separate validation set can be omitted, and cross-validation techniques can be employed. In this case, the dataset can be divided into a training set and a test set in an 8:2 ratio, resulting in 3073 training samples and 780 test samples.

D. Characteristics of the Dataset

1) Image Diversity: The images used to construct the dataset span various countries and regions, reflecting the sports popular in each region and the consequent variations in sports field styles and types. For instance, baseball is more popular in the USA, and thus, most samples include baseball diamonds. Football is more popular in Argentina, with most samples including football fields. In China, ground track fields often appear alongside football fields. These differences ensure regional diversity in the samples, which can support the detection of sports fields globally.

The images for the dataset are sourced from satellite platforms in China, France, the USA, and other countries, covering the mainstream stereo remote-sensing satellite platforms currently in operation. These platforms differ in orbital altitude (500–800 km), sensors carried, and image acquisition methods, and the obtained image data have different spatial resolutions (0.3–0.8 m), radiance resolutions, and imaging bands. These differences ensure diversity in image attributes, supporting comprehensive detection tasks involving the joint processing of multiple data types.

The images incorporated into the dataset include data captured across various seasons, times, and angles. Fig. 3 shows the numbers of samples obtained in different months. The images captured in winter exhibited snow interference in certain sports fields. The images captured in summer showed vegetation obscuring parts of the sports fields. Additionally, the images were captured during different times of the day, such as morning and afternoon, leading to diverse lighting conditions and shadow lengths, sizes, and directions. Finally, the angles of capture show variation, with some samples originating from forward-looking imagery and others from nadir-looking imagery, resulting in distinct textures, shadows, and spatial resolutions in images captured from different perspectives. These variations enrich the diversity of image information within the samples, thereby enhancing the robustness of the algorithms trained on this dataset.

2) Sample Diversity: The dataset comprises 3853 samples, which is sufficient to support the training of various large-scale object detection algorithms. Furthermore, the sample data are derived from actual remote-sensing imagery, encompassing urban, suburban, and rural scenarios (Fig. 4) and includes various land features, such as buildings, roads, farmlands, and vegetation. The sample scenes are complex, with numerous interfering elements, exhibiting the typical characteristics of complex and diverse scenarios found in remote-sensing datasets.

Because of the high orbital altitude of satellite platforms, the image acquisition process is inevitably affected by clouds and fog, resulting in cloud interference in multiple samples (Fig. 5). Additionally, because the imagery products are not regular rectangles, samples located at the edges of the imagery may suffer from information loss during cropping, with the missing parts typically filled in with black color; hence, these samples may contain black borders of varying sizes, shapes, and orientations (Fig. 5). These instances of information loss and cloud obstruction closely mimic real-world applications of remote-sensing data, further enriching the diversity of the samples and interfering with object detection.

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(a) (b) (c)

Fig. 4. Examples of sample data from different scenarios (a) urban sample, (b) suburban sample, and (c) rural sample.

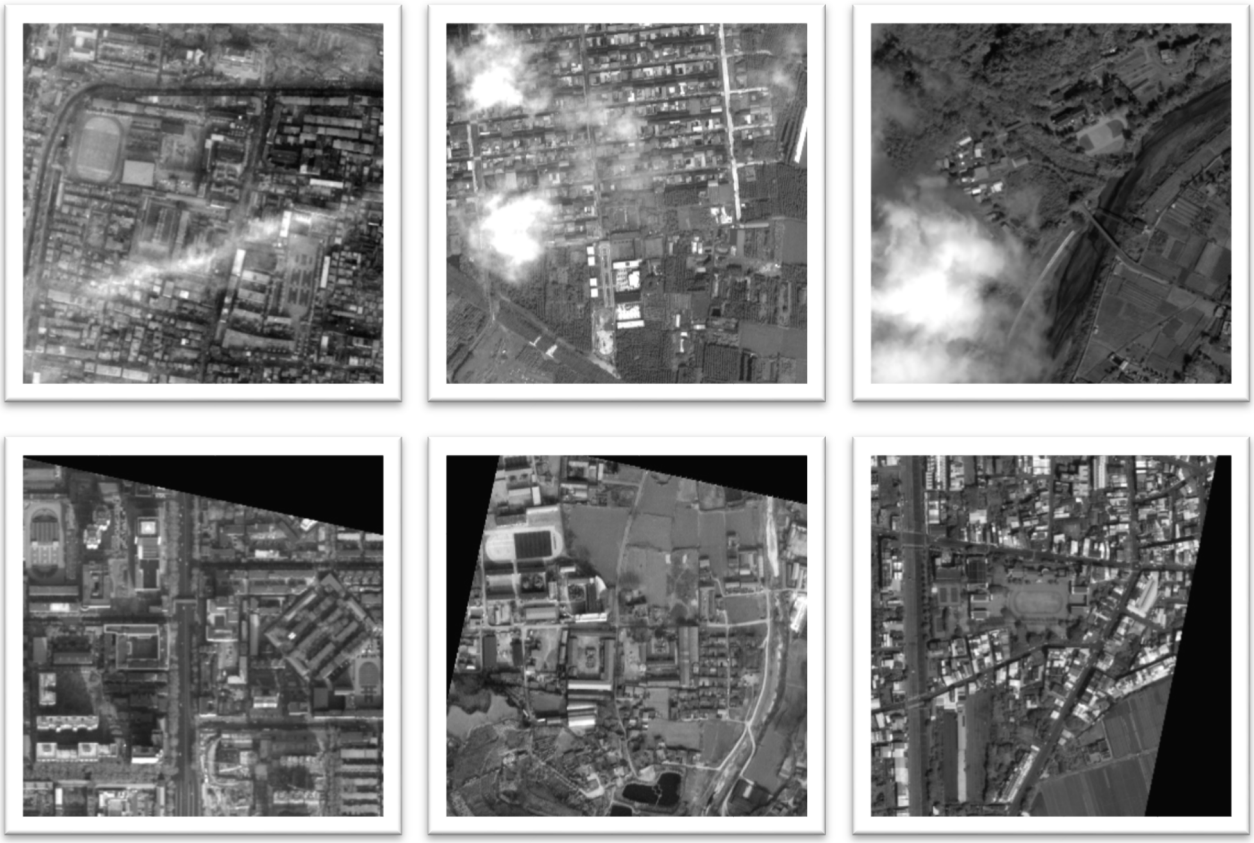


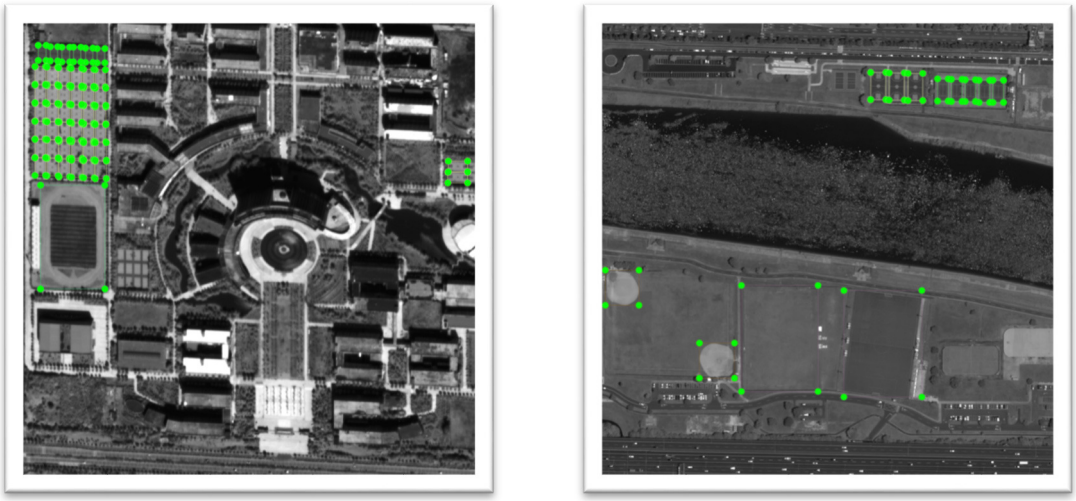
Fig. 5. Examples of special samples.

The sample data exhibit good informational diversity, with instances varying in type, number, distribution, and density. The highest number of instances in a sample is 45, as shown in Fig. 6(a), primarily featuring regularly arranged basketball courts and tennis courts with a high density; several samples include four or more types of instances, such as basketball courts, football fields, baseball diamonds, and tennis courts, as depicted in Fig. 6(b).

3) Adjustment Principle: The number of instances across different categories varies (Fig. 7). Basketball court instances are the most abundant, totaling 6136, whereas hockey field instances are the fewest, with only 173. This imbalance in instance distribution within the dataset is closely related to the actual distribution of sports fields; this characteristic poses a significant challenge to the stability of algorithms [64], particularly for categories with fewer instances, which are prone to accuracy fluctuations during training.

The primary factors affecting object detection are instance area and aspect ratio. As shown in Table III, area (average, AVG) varies significantly between different categories owing

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(a) (b)

Fig. 6. Typical sample examples. (a) Sample with a high number of instances and (b) sample with multiple instance types.

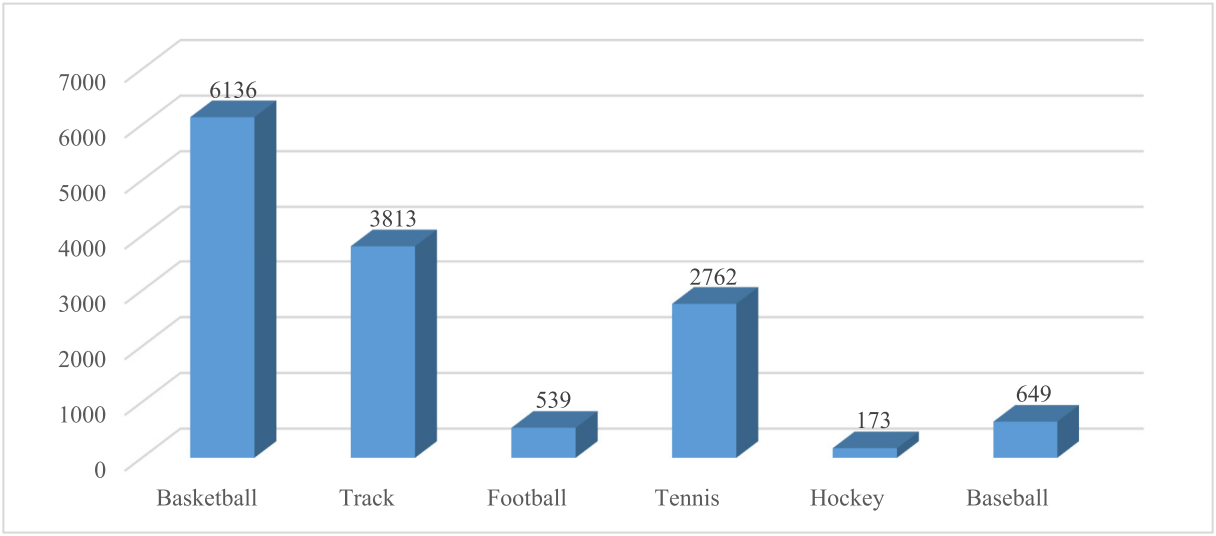


Fig. 7. Bar chart of instance counts for different categories.

TABLE III
STATISTICAL TABLE OF INSTANCE AREA INFORMATION BY CATEGORY (UNIT: SQUARE PIXELS)

Category	Basketball	Track	Football	Tennis	Hockey	Baseball
MAX	10,282	273,509	180,752	91,184	99,528	97,020
MIN	336	2730	1131	406	1830	2508
AVG	1908	25,977	16,459	1626	19,777	27,611
STD	1074	22,218	23,985	1068	24,968	21,518

to factors, such as differing spatial resolutions of the imagery and varying specifications of sports fields. Additionally, considerable variation exists in area among instances within the same category, as evidenced by the high variance values (standard deviation, STD) as well as the maximum (MAX) and minimum (MIN) values. According to the COCO dataset [65] standards, the proposed dataset contains 1455 small targets (area $< 32 \times 32$), accounting for approximately 10%, 8401 medium targets ($32 \times 32 <$ area $< 96 \times 96$), accounting for approximately 60%, and 4216 large targets (area $> 96 \times 96$), accounting for approximately 30% of the total. The aspect ratio of instances is also an important factor affecting object detection, particularly for anchor-based detection algorithms. The aspect ratios of instances in this dataset range from 0.24 to 3.16, with the specific distribution shown in Fig. 8.

The instance orientations in the proposed dataset, sourced from satellite imagery, are unaffected by gravity during the

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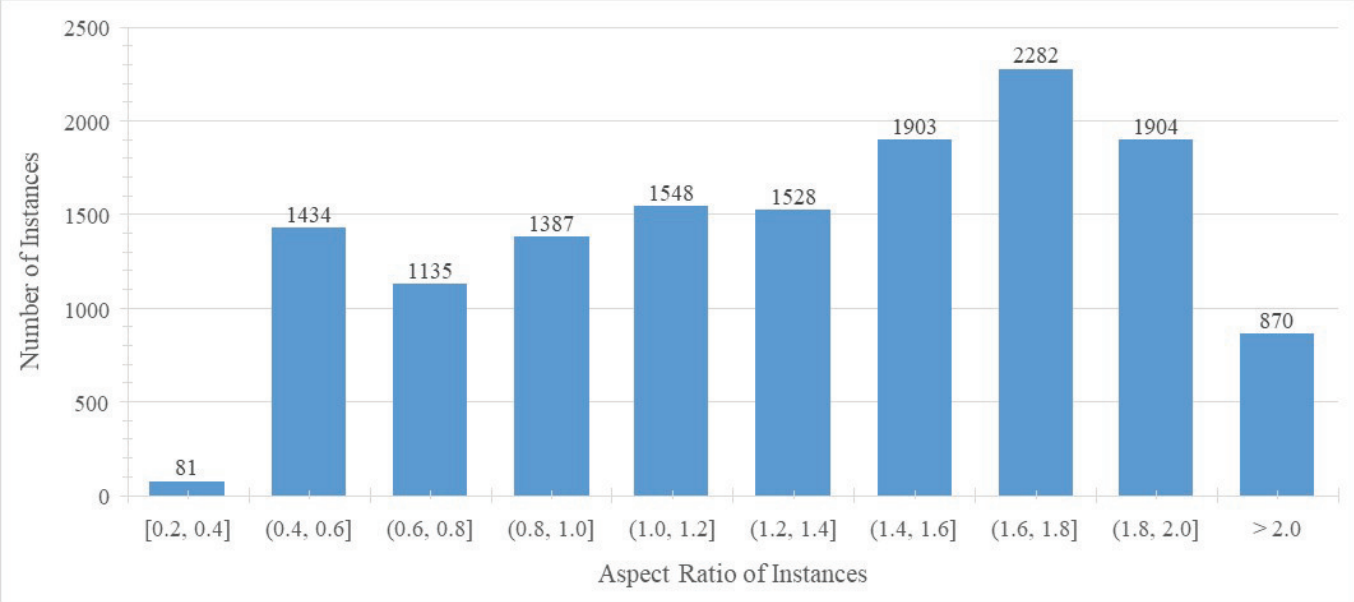


Fig. 8. Aspect ratio statistics of instances.

acquisition process. The instances have an equal probability of having any orientation within the range of $[-\pi, \pi]$, which is a characteristic of remote-sensing image datasets with diverse angles. Compared to datasets, such as face detection and pedestrian detection, where instance orientations are limited to the range $[-\pi/2, \pi/2]$, this dataset offers a richer and more varied range of orientation information. It can provide data support for research on rotation-invariant feature extraction and oriented target detection.

III. Benchmarks and Discussion

To provide a benchmark for subsequent researchers utilizing the proposed dataset, benchmark testing was conducted on several representative deep-learning-based object detection methods using SF_VHR_SSI. Classic algorithms, such as faster R-CNN [18], mask R-CNN [19], cascade R-CNN [20], Libra R-CNN [25], RetinaNet [36], CenterNet [38], and FCOS [39], were implemented on the MMDetection [66] platform. The dataset characteristics were analyzed in detail based on the detection results, laying the groundwork for future applications of the dataset. Furthermore, considering the advantages of the YOLO series algorithms in satellite remote-sensing image processing—such as rapid detection speed, high efficiency, adaptability to multiscale targets, and ongoing algorithmic improvements—more than 40 algorithmic models of YOLOv5 [30], YOLOv6 [31], YOLOv7 [34], YOLOv8 [32], YOLOv9 [33], YOLOv10 [35], and YOLOv11 [67] were implemented based on the Ultralytics platform. A detailed analysis of their results is provided to support the design of subsequent detection algorithms.

A. Evaluation Metrics

To assess the overall performance of the algorithm comprehensively, parameters are set to evaluate it from three perspectives: model size, detection efficiency, and detection accuracy. In terms of model size, the number of model parameters (Params) is used for evaluation. For detection efficiency, giga floating-point operations per second (GFLOPS) is employed as the metric, which represents the number of floating-point operations a model performs per second, scaled to billions (10^9 operations). Detection accuracy is assessed using a combination of metrics: mean average precision (mAP) at intersection over union (IoU) threshold 0.5 (mAP_50), mAP at IoU threshold 0.75 (mAP_75), mAP, mAP_small, mAP_medium, and mAP_large.

mAP_50 refers to the average of all class AP values (area under the precision/recall curve) at an IoU threshold of 0.5. The calculation method is detailed in [68]. Similarly, mAP_75 is the average of all class AP values at an IoU threshold of 0.75. mAP (IoU = 0.50:0.95) is the average of mAP values calculated at each 0.05 interval from 0.5 to 0.95 IoU thresholds, which includes the average of 10 mAP values, such as mAP_50, mAP_55, ..., mAP_95. mAP_small (IoU = 0.50:0.95) represents the average precision for small objects, which are typically defined as objects with an area less than 32 × 32 pixels [65]. Similarly, mAP_medium (IoU = 0.50:0.95) and mAP_large (IoU = 0.50:0.95) correspond to the average precision for medium and large objects, respectively, as categorized by the COCO dataset standards [65].

B. Classical Algorithms Experiment and Analysis

The hardware environment is outlined in Table II. All algorithms were implemented on this hardware platform. Considering the complexity of the object detection algorithm process and the numerous design choices and hyperparameter settings, all classical algorithms were uniformly implemented in the MMDetection platform for easy comparison between different methods. In terms of parameter settings, none of the algorithms used pretrained weight files, and the input image size was uniformly scaled to 640 × 640 pixels. The

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TABLE IV
STATISTICAL TABLE OF OVERALL DETECTION ACCURACY FOR CLASSICAL ALGORITHMS

Model	Backbone	Params ($M/10^6$)	GFLOPS	mAP	mAP				
					50	75	small	medium	large
Faster R-CNN	R-50	41.15	181.77	0.62	0.804	0.743	0.665	0.53	0.628
	R-101	60.14	242.5	0.628	0.799	0.751	0.67	0.54	0.638
Mask R-CNN	R-50	43.77	284.83	0.625	0.804	0.749	0.674	0.541	0.631
	R-101	62.76	345.57	0.626	0.79	0.759	0.65	0.532	0.647
Libra R-CNN	R-50	41.41	185.87	0.625	0.813	0.748	0.709	0.544	0.625
	R-101	60.4	246.6	0.625	0.801	0.754	0.677	0.533	0.63
Cascade R-CNN	R-50	68.94	237.36	0.644	0.813	0.759	0.681	0.552	0.655
	R-101	87.94	298.09	0.64	0.795	0.755	0.697	0.539	0.653
CenterNet	R-50	31.85	157.15	0.622	0.813	0.736	0.667	0.562	0.591
	R-101	50.79	217.89	0.628	0.817	0.743	0.666	0.551	0.621
RetinaNet	R-18	19.72	124.61	0.539	0.752	0.632	0.577	0.485	0.507
	R-50	36.21	164.84	0.581	0.778	0.692	0.586	0.506	0.592
	R-101	55.2	225.57	0.607	0.797	0.717	0.632	0.531	0.619
FCOS	R-50	31.85	157.19	0.61	0.822	0.72	0.629	0.544	0.613
	R-101	50.79	217.93	0.595	0.800	0.714	0.642	0.529	0.588

Note: R-18, R-50, and R-101 denote the ResNet-18, ResNet-50, and ResNet-101 networks, respectively [24].

training batch size was set to four or eight depending on the model size. Considering the slower convergence of one-stage algorithms owing to the absence of pretrained weights, the training period (epochs) was uniformly set to 36 and the learning rate was adjusted, with other hyperparameters uniformly adopting the "1×" training default parameters in MMDetection. Additionally, considering that FPNs have almost become standard features in the design of object detection algorithms, all algorithms have incorporated FPN modules. The specific detection results for various algorithms are shown in Table IV.

According to the overall mAP of the detection results, the cascade R-CNN series achieved the highest accuracy among the classical algorithms, reaching up to 64.4%. The RetinaNet series performed the worst, with the RetinaNet model based on the ResNet-50 backbone only achieving a detection accuracy of 58.1%. Overall, the accuracy of two-stage object detection algorithms generally surpassed that of one-stage object detection algorithms. The detection accuracy of the faster R-CNN algorithm was comparable to that of the CenterNet algorithm; however, the CenterNet algorithm exhibited smaller Params and GFLOPS metrics, indicating greater detection efficiency. Considering the GFLOPS metric, one-stage object detection algorithms were generally more efficient than two-stage object detection algorithms when using the same backbone network. Therefore, in general, two-stage object detection algorithms exhibited higher detection accuracy on the SF_VHR_SSI dataset and one-stage object detection algorithms offered higher processing efficiency. This finding was largely consistent with the theoretical analysis of deep-learning-based object detection algorithms, thereby validating the effectiveness and scientific validity of SF_VHR_SSI.

In terms of network depth, deeper backbone networks generally performed better in terms of accuracy on SF_VHR_SSI. For instance, models, such as faster R-CNN, mask R-CNN, CenterNet, and RetinaNet, achieved higher mAP and mAP_75 values when utilizing ResNet-101. The RetinaNet algorithm saw a corresponding increase in detection accuracy as the depth of its backbone network was increased incrementally from ResNet-18 to ResNet-50 and then to ResNet-101. However, network depth was not a panacea and deeper did not necessarily mean better. In the cascade R-CNN series of algorithms, the model based on ResNet-50 performed better in terms of accuracy. When ResNet-101 was used as the backbone network, a slight decrease in accuracy was observed, which was, of course, related to the complexity of the cascading structure of the cascade R-CNN and the size of the dataset. The performance of the Libra R-CNN network remained essentially consistent on both ResNet-50 and ResNet-100. These experimental results indicated that appropriately increasing network depth could improve the accuracy of detection algorithms on SF_VHR_SSI, which aligned with the fundamental theories of deep-learning-based object detection algorithms and further validated the effectiveness and scientific validity of SF_VHR_SSI.

When the detection accuracy in relation to target size within the SF_VHR_SSI dataset was examined, various algorithms achieved higher accuracy for small targets. This was partly owing to the high spatial resolution of the images in SF_VHR_SSI, where even the smallest sports field targets have 336 pixels (Table III). Moreover, the complexity of the background in remote-sensing imagery is one of the primary challenges in target detection and small targets have fewer background details within their bounding boxes,

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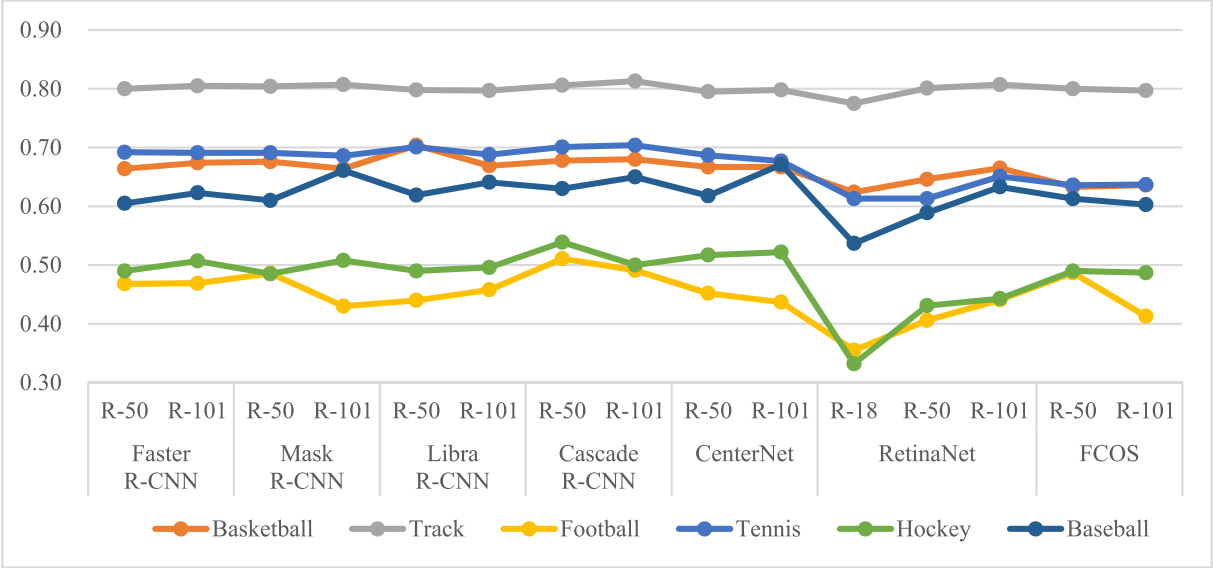


Fig. 9. Precision variation chart of various sports fields in different detection algorithms.

leading to fewer interference factors during feature extraction, reducing detection difficulty. By contrast, medium-sized sports fields posed the greatest detection challenge because of the substantial background information they contain. Particularly for medium-sized sports fields with a high aspect ratio, the number of background pixels within the bounding box can even exceed the number of target pixels. Coupled with the monotonous texture and limited spectral information in the original panchromatic satellite imagery, the detection accuracy for medium-sized sports fields was the lowest. As a typical remote-sensing image dataset, the complexity of the background in SF_VHR_SSI is one of the primary challenges for high-precision detection. The cascaded structure of cascade R-CNN has certain advantages in excluding background information (multiple structures progressively eliminate various background interferences [20]), which accounts for its superior detection accuracy. For future target detection based on SF_VHR_SSI, utilizing the contextual information and attention mechanisms effectively is crucial to eliminate background interference, thereby improving detection accuracy.

The SF_VHR_SSI dataset categorized sports fields into multiple classes, making it a fine-grained object detection dataset. The detection results of various classical algorithms for each category are presented in Table V. Overall, the accuracy trends of the classical algorithms across different categories were largely consistent (with similar detection accuracies for the same category across different algorithms), indicating that the category setup of the dataset is relatively reasonable, with clear interclass differences, ensuring the relative stability of detection accuracy for different algorithms across categories. Specifically, as shown in Fig. 9, the detection accuracy for ground track fields was the highest and most stable. This was because of the distinctive and prominent circular track structure of ground track fields, which provides a strong basis for detection, making it less challenging to identify. The detection accuracy for football fields was the lowest, with significant variations among different algorithms. There are four main reasons for this outcome: first, the standard football field covers a large area with complex backgrounds and numerous interfering elements; second, in addition to dedicated football fields, several multipurpose football fields exist that often appear in combination with ground track fields, posing a significant challenge to detection accuracy; third, football fields come in various sizes, including futsal fields, seven-a-side fields, and standard fields, with considerable variations in size and area, leading to large intraclass differences. Finally, futsal fields and seven-a-side football fields share certain similarities in area and shape with basketball courts and hockey fields and their texture is often uniform and close to that of the grass, making the detection process prone to interference from regular grasslands and other similar features. Therefore, future algorithm design should focus on the characteristics of football fields, adding modules to overcome errors caused by interclass similarities and intraclass variations. Additionally, the detection accuracy for hockey fields was relatively low, primarily because of fewer instances of hockey fields in the dataset (173). In the future, the detection accuracy for hockey fields can be improved by introducing weight allocation strategies to enhance the ability of the model to balance categories.

C. YOLO Series Algorithms Experiment and Analysis

All YOLO algorithms were implemented on the hardware platform shown in Table II. YOLOv7 and YOLOv9 (including GELAN and YOLOv9 algorithms) were realized based on the official code provided by the authors; the remaining algorithms were implemented using the Ultralytics software platform. In terms of parameter settings, none of the algorithms utilized pretrained weight files. The input image size was uniformly scaled to 640 $\times$ 640 pixels. The training period (epochs) was set to 300, and the training batch size was set to eight or 16 depending on the model size. All other hyperparameters were set to their default values according to the algorithms

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TABLE V
STATISTICAL TABLE OF OVERALL DETECTION ACCURACY FOR CLASSICAL ALGORITHMS

Model	Backbone	Params ($M/10^6$)	GFLOPS	mAP	mAP				
					50	75	small	medium	large
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	R-101	60.14	242.5	0.628	0.799	0.751	0.67	0.54	0.638
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	R-101	62.76	345.57	0.626	0.79	0.759	0.65	0.532	0.647
Libra R-CNN	R-50	41.41	185.87	0.625	0.813	0.748	0.709	0.544	0.625
	R-101	60.4	246.6	0.625	0.801	0.754	0.677	0.533	0.63
Cascade R-CNN	R-50	68.94	237.36	0.644	0.813	0.759	0.681	0.552	0.655
	R-101	87.94	298.09	0.64	0.795	0.755	0.697	0.539	0.653
CenterNet	R-50	31.85	157.15	0.622	0.813	0.736	0.667	0.562	0.591
	R-101	50.79	217.89	0.628	0.817	0.743	0.666	0.551	0.621
RetinaNet	R-18	19.72	124.61	0.539	0.752	0.632	0.577	0.485	0.507
	R-50	36.21	164.84	0.581	0.778	0.692	0.586	0.506	0.592
	R-101	55.2	225.57	0.607	0.797	0.717	0.632	0.531	0.619
FCOS	R-50	31.85	157.19	0.61	0.822	0.72	0.629	0.544	0.613
	R-101	50.79	217.93	0.595	0.800	0.714	0.642	0.529	0.588

Note: R-18, R-50, and R-101 denote the ResNet-18, ResNet-50, and ResNet-101 networks, respectively [24].

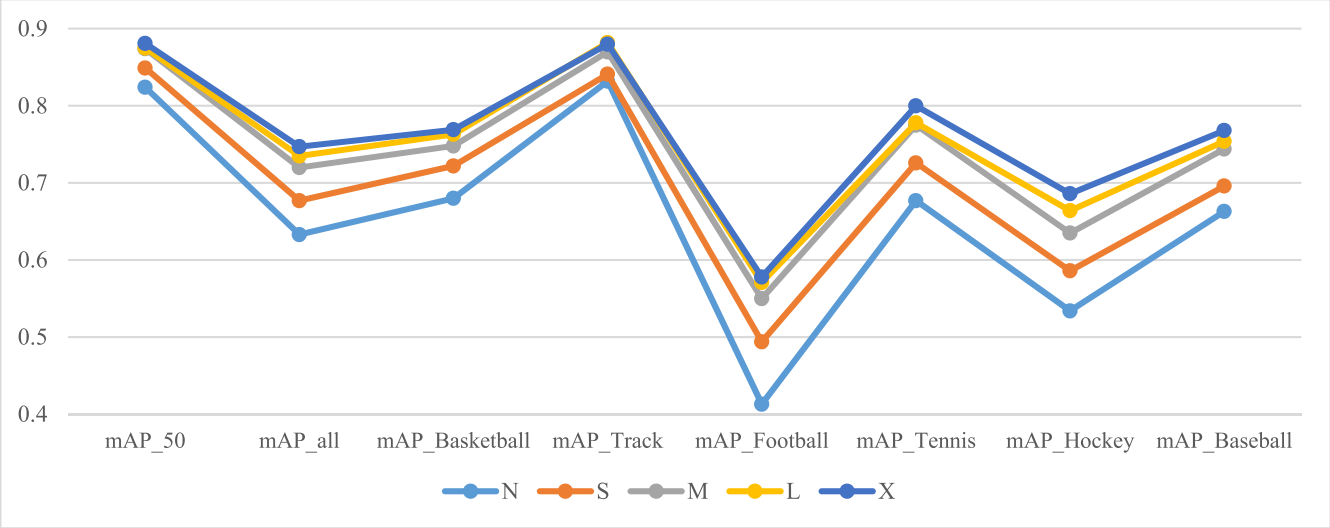


Fig. 10. Comparison of detection accuracy for different versions of the YOLOv5 model.

of the platform. As the most comprehensively updated object detection algorithms, the YOLO series incorporates a variety of new technologies, including more data augmentation techniques, attention mechanisms, and feature fusion modules. The overall detection results are superior to those of classical object detection algorithms, with specific detection outcomes presented in Table VI.

When examining individual algorithms, such as YOLOv5, consistent improvement was observed in the overall detection accuracy on the SF_VHR_SSI dataset, as well as the accuracy for individual categories, with the increase in depth and complexity of the model architecture (Fig. 10). However, deeper and more complex models do not always yield better detection accuracy. For instance, the L version of YOLOv10 outperformed the larger X version in detection accuracy. This aligns with the conclusions of classical detection algorithms and reaffirms the effectiveness and scientific validity of SF_VHR_SSI. Significant enhancement was observed in various accuracy metrics (mAP) when transitioning from the smaller versions (N and S) to the medium version (M); however, the improvement in accuracy metrics from the medium version to the larger versions (L and X) was less pronounced and the efficiency metrics (GFLOPS) significantly decreased. This correlation is attributed to the size of SF_VHR_SSI and the complexity of the models, offering guidance for future model version design based on SF_VHR_SSI.

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TABLE VI
YOLO SERIES ALGORITHM DETECTION ACCURACY STATISTICS TABLE

Model	Version	Params	GFLOPS	mAP_50	mAP	mAP					
						Basketball	Track	Football	Tennis	Hockey	Baseball
v5	N	2.51	7.20	0.824	0.633	0.680	0.832	0.413	0.677	0.534	0.663
	S	9.12	24.10	0.849	0.677	0.722	0.841	0.494	0.726	0.586	0.696
	M	25.07	64.40	0.874	0.720	0.748	0.870	0.550	0.775	0.635	0.744
	L	53.17	135.30	0.875	0.735	0.763	0.882	0.571	0.778	0.664	0.754
	X	97.21	246.90	0.881	0.747	0.769	0.880	0.578	0.800	0.686	0.768
v6	N	4.24	11.90	0.805	0.630	0.690	0.842	0.402	0.680	0.490	0.675
	S	16.31	44.20	0.836	0.684	0.730	0.855	0.507	0.739	0.576	0.698
	M	52.00	161.60	0.831	0.692	0.745	0.866	0.514	0.765	0.528	0.736
	L	110.90	391.90	0.843	0.696	0.756	0.872	0.510	0.772	0.535	0.730
	X	173.03	611.20	0.858	0.723	0.756	0.882	0.528	0.780	0.675	0.713
v7	T	6.03	13.20	0.819	0.623	0.670	0.819	0.438	0.666	0.472	0.674
	-	37.22	105.20	0.874	0.712	0.751	0.860	0.542	0.764	0.610	0.744
	X	70.85	189.00	0.884	0.724	0.761	0.862	0.558	0.761	0.664	0.738
v8	N	3.01	8.20	0.832	0.645	0.691	0.836	0.440	0.677	0.537	0.687
	S	11.14	28.70	0.849	0.691	0.737	0.858	0.492	0.746	0.587	0.726
	M	25.86	79.10	0.877	0.724	0.754	0.879	0.558	0.777	0.656	0.722
	L	43.63	165.40	0.865	0.731	0.765	0.885	0.554	0.783	0.631	0.766
	X	68.16	258.10	0.878	0.739	0.770	0.884	0.580	0.790	0.643	0.765
GELAN	T	1.91	7.30	0.859	0.683	0.689	0.845	0.485	0.684	0.676	0.720
	S	7.20	26.90	0.888	0.739	0.750	0.874	0.548	0.766	0.734	0.759
	M	20.07	77.10	0.908	0.768	0.775	0.883	0.639	0.802	0.725	0.784
	C	25.44	103.20	0.915	0.778	0.785	0.892	0.633	0.802	0.765	0.793
	E	58.06	192.20	0.914	0.779	0.789	0.893	0.608	0.808	0.803	0.769
v9	T	2.66	11.00	0.857	0.689	0.701	0.850	0.508	0.701	0.628	0.745
	S	9.75	39.60	0.887	0.733	0.742	0.871	0.574	0.763	0.678	0.767
	M	32.77	132.40	0.910	0.774	0.783	0.890	0.633	0.809	0.735	0.793
	C	51.01	238.90	0.910	0.769	0.783	0.894	0.613	0.812	0.744	0.766
	E	69.42	244.90	0.914	0.781	0.793	0.894	0.623	0.810	0.792	0.776
v10	N	2.71	8.40	0.812	0.641	0.685	0.831	0.435	0.702	0.523	0.668
	S	8.07	24.80	0.847	0.679	0.720	0.838	0.483	0.746	0.573	0.712
	M	16.49	64.00	0.870	0.714	0.744	0.862	0.525	0.767	0.645	0.741
	B	20.46	98.70	0.869	0.711	0.743	0.856	0.548	0.767	0.634	0.716
	L	25.77	127.20	0.867	0.730	0.750	0.867	0.575	0.789	0.635	0.764
	X	31.67	171.10	0.852	0.719	0.750	0.874	0.544	0.788	0.635	0.724
v11	N	2.59	6.40	0.814	0.632	0.668	0.840	0.429	0.686	0.491	0.680
	S	9.43	21.60	0.851	0.683	0.724	0.856	0.487	0.744	0.570	0.718
	M	20.06	68.00	0.873	0.723	0.753	0.873	0.537	0.779	0.660	0.734
	L	25.32	87.30	0.872	0.734	0.767	0.884	0.529	0.786	0.685	0.754
	X	56.88	195.50	0.890	0.756	0.771	0.894	0.589	0.808	0.706	0.766

Note: version "N" denotes "Nano," which is the smallest model within the YOLO family; "S" stands for "Small," indicating the relatively small model in the series; "M" represents "Medium," referring to the model of medium size; "L" denotes "Large," the larger model in the series; and "X" corresponds to "Extra-large," the largest model in the YOLO family. "T" represents "Tiny," which is the smallest lightweight model in the YOLOv7 and YOLOv9 algorithms; "C" denotes "Compressed," which refers to the larger compressed model in the YOLOv9 algorithm; and "E" stands for "Extended," which is the largest extended model in the YOLOv9 algorithm. Finally, "B" denotes "Balance," representing the balanced model within the YOLOv10 series.

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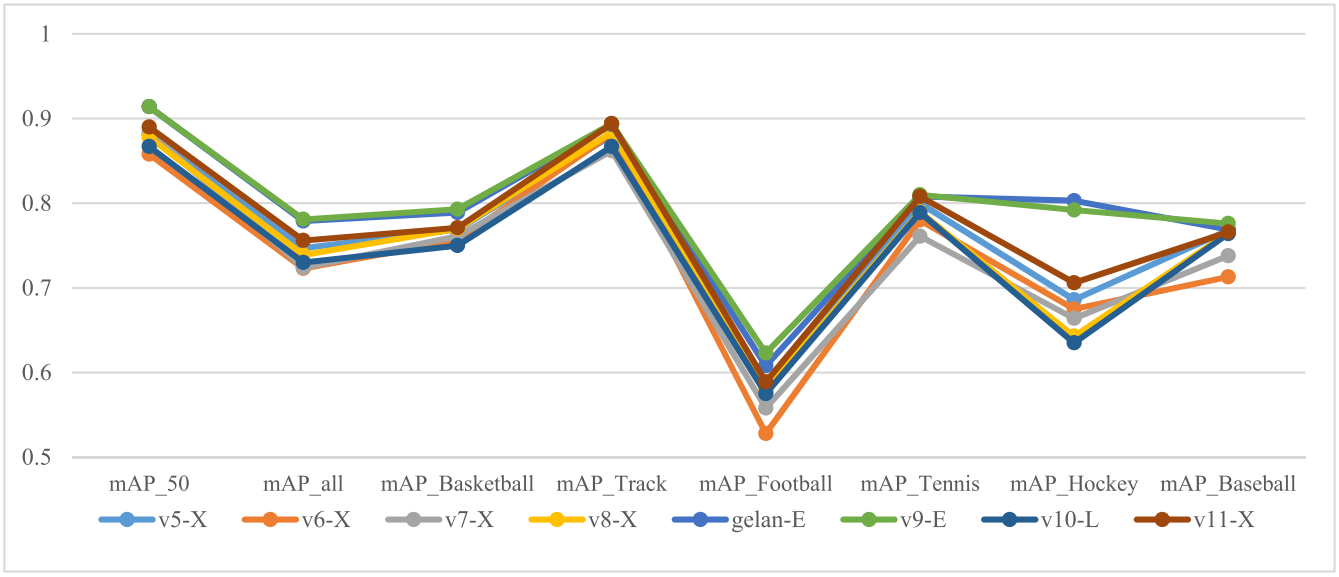


Fig. 11. Comparison of detection accuracies for optimal versions of YOLO series algorithms.

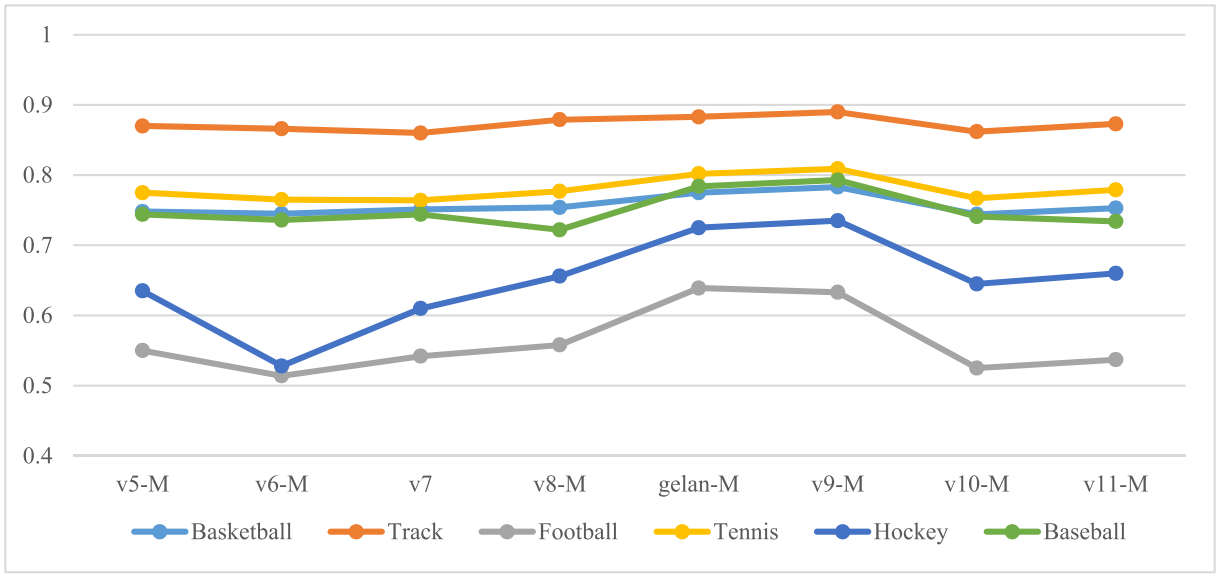


Fig. 12. Comparison of detection accuracies of YOLO series algorithm M versions across subcategories.

In the context of overall detection accuracy, YOLOv9 and the GELAN algorithm within the YOLO series exhibited superior performance on the SF_VHR_SSI dataset. This superior performance is attributed to their backbone networks, which utilize the GELAN backbone network that integrates the advantages of the cross stage partial network (CSPNet) [69] and efficient layer aggregation network (ELAN) [70]. YOLOv9 added an auxiliary reversible branch [33] on the basis of the GELAN network. However, from the perspective of detection accuracy, the improvement brought by the auxiliary reversible branch was limited, with a decrease observed in the C version. Comparing the optimal versions of each category of algorithms (Fig. 11) revealed that the greatest advantage of YOLOv9 and the GELAN algorithm was their high detection accuracy for hockey fields, which had the fewest instances in the dataset. This indicates that the GELAN network can handle class imbalance issues and be employed as a backbone network for "long-tail" datasets. Among all the optimal versions, YOLOv10-L had the highest efficiency but low accuracy, with an mAP_all of only 73%. GELAN-E ranked second in efficiency and possessed a higher accuracy, with an mAP_all value reaching 77.9%. Furthermore, YOLOv11-X had an efficiency comparable to that of GELAN-E and exhibited a high detection accuracy of 75.6%. The GELAN network achieved a good balance between efficiency and accuracy on SF_VHR_SSI, offered the best detection performance, and could serve as a benchmark for subsequent algorithm optimization.

In the analysis of the M versions of various algorithms, the detection outcomes of the YOLO series algorithms across different subcategories were examined (Fig. 12). Consistent with the detection results of traditional algorithms, track

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fields, because of their distinctive structural features, had the highest and most consistent detection accuracy. Tennis courts, owing to their smaller footprint and minimal interference from background elements, exhibited detection precision and stability that ranked just below those of track fields. The detection accuracy for baseball and basketball courts was slightly lower than that for tennis courts. Hockey fields, owing to their limited instances, displayed lower detection accuracy and stability. Football fields, impacted by substantial background interference, significant intraclass variation, simplicity in structure, and monotonous texture, had the lowest detection accuracy. The concordance between the detection results of the YOLO series algorithms and those of traditional algorithms reaffirmed the validity and robustness of the SF_VHR_SSI dataset. Furthermore, within the YOLO series, YOLOv9 and the GELAN algorithm were more adept at addressing the class imbalance of the dataset, particularly achieving higher precision in the detection of hockey fields; however, the detection accuracy for football fields remained low. Consequently, the targeted integration of attention mechanisms and other innovative technologies to enhance the detection accuracy for football fields, without compromising the accuracy of other categories, should be a priority in the design of algorithms based on SF_VHR_SSI.

IV. SF_GELAN and Experiments

Incorporating the characteristics of very-high-resolution imagery and the richness and diversity of samples from the SF_VHR_SSI dataset (Section II), the SF_GELAN network was developed based on the GELAN [33] framework within the YOLO series to address the challenges posed by class imbalance and the complexity in football field detection (Section III). Experimental ablation studies were conducted on the critical modules of the SF_GELAN network, with GELAN-C serving as the baseline. Furthermore, comparisons were made between different versions of the SF_GELAN network and the latest iterations of the YOLO algorithms, including YOLOv8, GELAN, YOLOv9, YOLOv10, and YOLOv11. Finally, the universality of the SF_GELAN network was validated by conducting experiments on the classic remote-sensing image dataset DIOR using the SF_GELAN-S algorithm.

A. SF_GELAN Network

In the SF_GELAN network, a FasterCSPELAN4 feature extraction module was designed by integrating PConv [56] with the FasterNet Block [56] and RepNCSPELAN4 [33] modules. This module was applied throughout the network to enhance feature extraction capabilities while optimizing detection efficiency, overcoming the interference of complex background information in remote-sensing imagery and improving the detection accuracy of targets. In the neck region of the network, an SE mechanism was introduced, and a squeeze–excitation attention (SEAttention) mechanism was designed to enhance the feature representation ability and improve the detection accuracy of hockey fields. Additionally, by combining FasterCSPELAN4, the detection accuracy of football fields was further enhanced. In the post-processing stage, Merge-NMS was employed to fuse multiple high-precision detection boxes, further improving the overall detection accuracy of the network. Moreover, the SF_GELAN network supports model scaling. Using model scaling techniques, three versions of the SF_GELAN model were created—small (SF_GELAN-S), medium (SF_GELAN-M), and large (SF_GELAN-B)—for different scenarios.

1) FasterCSPELAN4 Module: RepNCSPELAN4 is an important component of the GELAN network that is primarily responsible for extracting features from input images. It achieves feature fusion through operations, such as channel splitting and merging, providing a rich set of feature information to ensure detection accuracy. The RepNCSPELAN4 module enhances computational efficiency by optimizing convolution operations, reducing redundant calculations, and optimizing memory access patterns while maintaining high accuracy. It follows the modular design philosophy of the YOLO series, offering excellent independence and scalability. It can be integrated easily into different versions of YOLOv9 and can be extended or modified as needed to accommodate specific detection tasks. The overall structure of the RepNCSPELAN4 module combines CSP modules [69] and ELAN modules [70], as shown in Fig. 13(a). RepNCSP consists of convolutions (Conv) and a variable number of RepNBottleneck modules, the detailed structure of which is shown in Fig. 13(b). Their number is determined by the model width factor N. RepNBottleneck is a basic module with a residual structure that incorporates the RepConv structure [71], composed of multiple Convs with different kernels, the detailed structure of which is shown in Fig. 13(c). It is responsible for feature extraction, compression, and fusion and utilizes structural reparameterization for lightweight deployment. However, the reparameterizable characteristic increases the complexity of model implementation and the rigidity of model deployment. Additionally, during training, the parameters of the two Convs within the RepConv structure must be adjusted carefully to balance model complexity and performance.

In response to the aforementioned issues and inspired by FasterNet [56], PConv was introduced to integrate with the FasterNet Block to redesign the RepNBottleneck module into FasterBottleneck. FasterBottleneck retains the residual structure of RepNBottleneck, as illustrated in Fig. 14.

The core concept of the FasterNet Block module is as follows. Initially, PConv is utilized to process the input data, applying conventional convolution to some of the input channels while leaving the rest unchanged. This design of PConv reduces computational redundancy and the number of memory accesses, enhancing computational efficiency. Subsequently, two consecutive pointwise convolutions (PWConvs) are employed to construct a simple multilayer perceptron (MLP) for feature transformation, aggregating feature information from PConv. This enables the network to utilize information from all channels more effectively and simultaneously enhances the focus on central positions [56]. Furthermore, to increase the stability of the network while maintaining the diversity of features, a batch normalization layer and an activation layer, the rectified linear unit (ReLU), are inserted between the two PWConvs. Then, DropPath is applied to regularize the output of the MLP, improving

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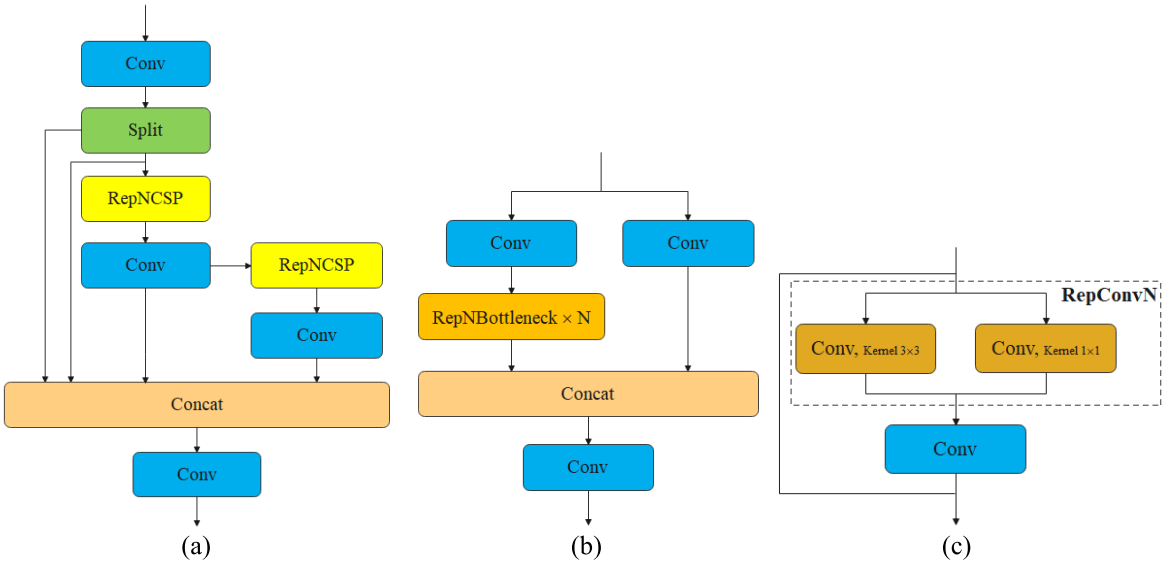


Fig. 13. RepNCSPELAN4 module structure. (a) RepNCSPELAN4, (b) RepNCSP, and (c) RepNBottleneck.

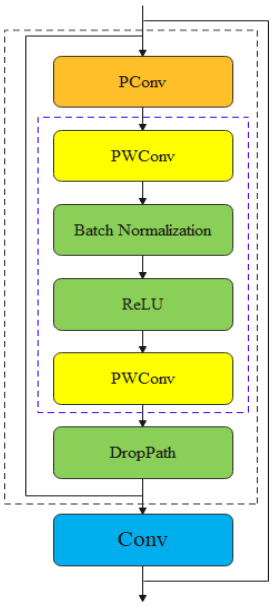


Fig. 14. FasterBottleneck module structure (the FasterNet Block module is enclosed within the black dashed line, and the MLP within the FasterNet Block module is enclosed within the blue dashed line).

network generalization while maintaining high computational efficiency. Finally, the regularized output is added to the original input to form a residual connection, further enhancing the feature representation and generalization capabilities of the network.

The FasterBottleneck module enhances computational and memory access efficiency by employing PConv and PWConvs, thereby ensuring the efficiency of the network. Using meticulously designed feature aggregation and shortcut connections, the accuracy of the network is ensured. In terms of overall structure, it is identical to RepNBottleneck, possessing good independence and scalability. By integrating the FasterBottleneck module into CSP and ELAN structures, a more efficient, accurate, and independent and scalable FasterCSPELAN4 module is obtained, which is used to improve the precision and efficiency of the SF_GELAN network.

2) SEAttention Module: The SE mechanism was first proposed in [72]. Its advantage lies in explicitly modeling and adjusting the channel features within convolutional neural networks to enhance the representation of useful features and suppress less important ones. The squeeze operation is primarily achieved via global average pooling, which aggregates global spatial information for each channel, enabling the network to leverage global contextual information. Following aggregation, the excitation operation is primarily implemented based on two fully connected layers and a sigmoid function. This process encodes the global information obtained from the squeeze operation into channelwise weights, which are then multiplied with each channel of the original feature map to amplify important features and suppress less significant ones. Finally, residual connections are added to improve gradient flow within the network, mitigating the issues of vanishing or exploding gradients. This mechanism is designed simply, focusing on adjusting channel weights to highlight features without additional network branches or complex operations. The entire module has a small number of parameters and belongs to a lightweight channel attention mechanism with good flexibility and general applicability, making it easy to integrate into various CNN architectures. Subsequently, the fully connected layers in the SE mechanism were explained in [57], introducing a multibranch fully connected layer structure (aggregated MLP) and enhancing the ability of the network to learn global features.

In this study, the SE mechanism is introduced primarily to highlight the features of sports field targets via their weight information while suppressing the complex background information in remote-sensing images, thereby improving the detection accuracy for sports fields. The specific implementation approach is as follows. First, global average pooling is applied to the input feature map X to perform the squeeze

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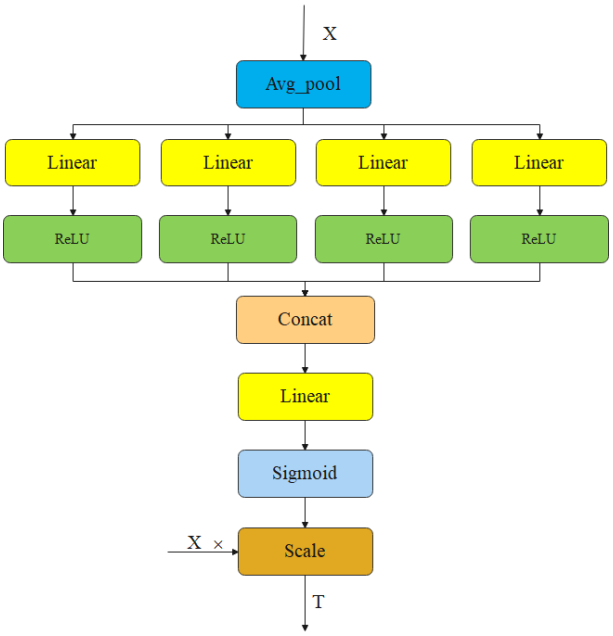


Fig. 15. SEAttention module structure (the hyperparameter cardinality is set to four, the letter X denotes the input feature map, and the letter T represents the output feature map after processing through SEAttention).

operation, compressing the spatial dimensions (height and width) into statistical information along the channel dimension, providing global contextual information for subsequent channel relationship modeling. Second, a multibranch fully connected layer structure [57] (where the number of branches, cardinality, is a hyperparameter of SEAttention) is introduced to process the global information output from the squeeze operation in parallel. Each fully connected layer reduces the dimensionality of the feature map from the input value (in_channel) to in_channel divided by the reduction ratio (reduction, which is a hyperparameter of SEAttention) and then introduces nonlinearity using the ReLU activation function, enhancing the expressive power of the model. Subsequently, all feature maps output from the multibranch structure are concatenated and fed into a new fully connected layer for processing and channel weights are generated via the sigmoid function to adjust the importance of each channel in the original feature map, enhancing useful features and suppressing less important ones. Finally, the original input feature map X is multiplied by the generated channel weights to achieve feature recalibration, enabling the network to adjust the response of the feature map adaptively, improving the feature representation and network performance.

The structure of SEAttention is shown in Fig. 15. Overall, the design of SEAttention is simple. The squeeze operation is completed via simple global average pooling without involving weight calculations, resulting in very low computational costs. In the excitation operation, although fully connected layers with higher computational costs are introduced, the computational complexity is controlled via the design of dimensionality reduction (the reduction parameter) and grouping (the cardinality parameter). Consequently, by appropriately configuring the two hyperparameters, reduction and cardinality, an effective balance between the efficiency and performance of SEAttention can be achieved. The subsequent experimental section provides a detailed analysis of these two hyperparameters.

3) Merge-NMS Technology: Merge-NMS is a natural extension of NMS. NMS is a widely used technique in object detection tasks for eliminating overlapping bounding boxes. Merge-NMS is a method employed when processing multiscale feature maps to merge and optimize detection results from different scales. This concept has been adopted in the design and optimization of object detection models in numerous studies [18], [29], [30]. Overall, Merge-NMS is an effective postprocessing technique in object detection that enhances accuracy and robustness by merging detection results from feature maps of varying scales, particularly excelling in handling multiscale targets. However, careful tuning of the NMS threshold and fusion strategies is necessary in practical applications to achieve optimal detection outcomes.

The fusion strategy is crucial for the effectiveness of Merge-NMS technology. Common fusion strategies include MAX score fusion, average score fusion, and weighted score fusion. Weighted score fusion can be further categorized based on the method of weight assignment into confidence-weighted fusion, IoU-weighted fusion (with various methods for calculating IoU [49], [50], [51], [52]), and combined weighted fusion. In this study, the adopted fusion strategy combines confidence and IoU weight calculation. The specific calculation method is as follows. Initially, the IoU values between all candidate bounding boxes are calculated (generating an IoU matrix) to identify the boxes that overlap and require merging. To reduce computational load, the simplest IoU ratio is used without incorporating additional constraints, such as angles, distances, or shapes. Subsequently, the IoU values are multiplied by the confidence scores to obtain weight values; the magnitude of these weights reflects their contribution to the final merged bounding box. Then, the coordinates of all candidate bounding boxes are weighted averaged using the calculated weights to obtain the coordinates of the merged bounding box. This spatially fuses multiple candidate boxes, enabling the final box coordinates to improve the representation of the location of the overlapping boxes. Finally, the MAX value among the class scores of each candidate bounding box is calculated and the corresponding class is selected as the category of the bounding box.

B. Experiments

Based on the hardware and software conditions listed in Table II, an experimental platform was established to conduct ablation, comparative, and generalization experiments. For the ablation experiments, the focus was on the placement of the SEAttention module within the network, the two hyperparameters of the SEAttention module, and the detection performance of the FasterCSPELAN4, SEAttention, and Merge-NMS modules. In the comparative experiments, different versions of the SF_GELAN network were compared with corresponding versions of algorithms, such as YOLOv8, YOLOv9, YOLOv10, and YOLOv11. In generalization experiments, the performances of different algorithms (small version)

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TABLE VII
DETECTION RESULT STATISTICS FOR SEATTENTION MODULE ADDED AT DIFFERENT NETWORK LOCATIONS

Scheme	Params	GFLOPS	mAP_50	mAP	mAP					
					Basketball	Track	Football	Tennis	Hockey	Baseball
Baseline	25.442	103.20	0.915	0.778	0.785	0.892	0.633	0.802	0.765	0.793
One	25.507	103.20	0.917	0.780	0.776	0.887	0.612	0.796	0.814	0.794
Two	25.589	103.30	0.915	0.783	0.783	0.889	0.619	0.803	0.811	0.795
Three	25.655	103.40	0.923	0.779	0.782	0.882	0.620	0.804	0.794	0.789

on the classic DIOR remote-sensing image object detection dataset were analyzed.

Regarding the evaluation metrics, a comprehensive assessment of the algorithms was performed based on model parameter counts, model detection efficiency (GFLOPS), and model detection accuracy (mAP). In terms of parameter settings, no pretrained weight files were used in any of the experiments. The input image size was uniformly scaled to 640 × 640 pixels. The number of training epochs was set to 300, and the training batch size was set to 16. All other hyperparameters were set to the default values of the algorithms of the platform.

1) Ablation Study of the SEAttention Module: The SEAttention module was subjected to ablation experiments focusing primarily on three aspects: the location of addition, the number of branches (cardinality), and the reduction ratio (reduction). The location of addition was investigated to understand the impact of SEAttention at different locations within the model on detection performance, providing insights for the subsequent integration of similar attention mechanisms. In line with the experience in model construction and the goal of designing efficient and lightweight networks, three experimental schemes were devised for the location of addition.

1) Scheme 1 involved adding the SEAttention module at the end of the backbone network, with the expectation that it would optimize the features extracted by the backbone, enhance target feature information, suppress background noise, and provide high-quality feature information for the subsequent feature fusion in the neck of the network.
2) Scheme 2 incorporated the SEAttention module on the small, medium, and large object detection feature maps in the neck of the network, aiming to improve the quality of these feature maps further and supply high-quality feature information to the detection head.
3) Scheme 3 combined the additions from both previous schemes, placing the SEAttention module both at the end of the backbone and on the feature maps in the neck to observe its effect on the overall detection performance of the network.

In the experiments, the cardinality of the SEAttention module was set to four and the reduction ratio was set to 32 [57]. The specific detection results for the three experimental schemes are presented in Table VII.

The SEAttention module has a minimal number of parameters and exerts a negligible impact on detection efficiency (GFLOPS). All three addition schemes led to improvements in precision. For the mAP_50 T, Scheme 3 achieved the highest detection accuracy, with an increase of 0.8% compared with that of the baseline. For the mAP metric, Scheme 2 demonstrated the highest detection accuracy, showing an improvement of 0.5% relative to the baseline. When considering the precisions of specific categories, the addition of SEAttention had a noticeable effect on the detection accuracy for football fields but significantly enhanced the detection accuracy for hockey fields. This indicates that the channel modeling and adjustment capabilities of the SEAttention module can mitigate the impact of class imbalance in the dataset. However, its compression operation can lead to some information loss, resulting in a decrease in detection accuracy for targets with high intraclass variability. Considering both detection efficiency and accuracy, this study selected Scheme 2 for the subsequent experiments and for the integration of the SEAttention module within the SF_GELAN network.

Regarding the reduction ratio, based on the experiences from [57] and [72] and the concept of adaptive adjustment, five experimental schemes were established. For detection experiments, Scheme 1 sets the reduction ratio to a fixed value of 16 [72], Scheme 2 sets the reduction ratio to a fixed value of 32 [57], and Scheme 3 sets the reduction ratio to a fixed value of 64. Scheme 4 correlated positively with the number of input channels (the larger the number of channels, the greater the reduction ratio), adaptively adjusting the value of reduction; this hypothesis posits that the larger the number of channels, the higher the redundancy among channel features, necessitating a larger compression ratio to extract useful information. Scheme 5 correlated negatively with the number of input channels (the larger the number of channels, the smaller the reduction ratio), adaptively adjusting the value of reduction; this hypothesis is based on the observation that feature maps for medium and large objects often have a higher number of channels (such as 512 and 640) and are located at the end of the FPN structure, having undergone multiple levels of extraction and summarization, thus requiring a smaller compression ratio to maintain the integrity of the information.

During the experiments, the cardinality of the SEAttention mechanism was set to four. The specific detection results for the five experimental schemes are detailed in Table VIII. A smaller reduction ratio led to a larger number of parameters and consequently a lower detection efficiency. However, overall, SEAttention is considered a lightweight attention mechanism, and changes in the reduction ratio had a minimal impact on efficiency. Regarding accuracy metrics, Scheme 1 (reduction = 16) failed to capture the relationships between channels effectively, resulting in lower detection accuracy. For the mAP metric, Scheme 2 demonstrated the best detection

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TABLE VIII
DETECTION RESULT STATISTICS FOR THE SEATTENTION MODULE UNDER DIFFERENT REDUCTION RATIOS

Scheme	Params	GFLOPS	mAP_50	mAP	mAP					
					Basketball	Track	Football	Tennis	Hockey	Baseball
Baseline	25.442	103.20	0.915	0.778	0.785	0.892	0.633	0.802	0.765	0.793
1	25.740	103.40	0.911	0.772	0.776	0.885	0.618	0.801	0.776	0.774
2	25.589	103.30	0.915	0.783	0.783	0.889	0.619	0.803	0.811	0.795
3	25.515	103.30	0.907	0.775	0.782	0.892	0.605	0.811	0.789	0.769
4	25.606	103.30	0.915	0.780	0.782	0.888	0.632	0.802	0.808	0.768
5	25.720	103.40	0.916	0.776	0.779	0.890	0.608	0.805	0.791	0.782

TABLE IX
DETECTION RESULT STATISTICS FOR THE SEATTENTION MODULE WITH DIFFERENT BRANCH NUMBERS

Cardinality	Params	GFLOPS	mAP_50	mAP	mAP					
					Basketball	Track	Football	Tennis	Hockey	Baseball
Baseline	25.442	103.20	0.915	0.778	0.785	0.892	0.633	0.802	0.765	0.793
2	25.515	103.30	0.918	0.769	0.776	0.882	0.616	0.800	0.753	0.786
3	25.552	103.30	0.905	0.765	0.775	0.883	0.597	0.809	0.739	0.785
4	25.589	103.30	0.915	0.783	0.783	0.889	0.619	0.803	0.811	0.795
5	25.626	103.30	0.914	0.777	0.781	0.890	0.624	0.814	0.774	0.780
6	25.663	103.40	0.909	0.771	0.783	0.885	0.604	0.796	0.781	0.778
8	25.737	103.40	0.900	0.766	0.782	0.890	0.596	0.805	0.740	0.784

accuracy, indicating that its reduction ratio was appropriately sized and essentially achieved the expected effects of the attention mechanism. Scheme 3, with a larger reduction ratio (reduction = 64), suffered from some loss of detail features, leading to lower detection accuracy. Compared with the baseline algorithm, Scheme 4, which adopted the concept of a reduction ratio positively correlated with the number of input channels, also achieved good detection results. This scheme showed better detection accuracy for football fields. Scheme 5, which hypothesized a reduction ratio negatively correlated with the number of input channels, maintained comparable overall accuracy to the baseline experiments. In conclusion, a fixed value of 32 was selected as the reduction ratio for the SEAttention module for the subsequent experiments.

In the SEAttention module, the value of cardinality determines the number of parallel fully connected layers. Increasing the cardinality can enhance the expressive power of the model because it enables it to capture information in different subspaces. However, this also implies an increase in the number of model parameters and computational complexity. To analyze the impact of the numbers of branches on model performance, experiments were conducted with cardinality set to two, three, four, five, six, and eight. The detection results are presented in Table IX. The detection results show that the more the branches, the greater the number of parameters and the lower the detection efficiency. Nevertheless, because SEAttention is a lightweight attention mechanism, the overall change in detection efficiency is not significant. In terms of detection accuracy, the model with two branches achieved the highest mAP_50 value and the model with four branches achieved the highest mAP value. The model with five branches showed detection accuracy comparable to that of the baseline model, particularly with higher detection accuracy for football fields. Based on the mAP metric as the primary criterion, in this study, the number of branches for the SEAttention module in subsequent experiments was set to four.

2) Ablation Studies of the SF_GELAN Network: Ablation experiments of the SF_GELAN network were primarily conducted to analyze and validate the effectiveness of the FasterCSPELAN4 feature extraction module, the SEAttention mechanism, and the Merge-NMS postprocessing technique on the SF_VHR_SSI dataset. To observe and analyze the specific impacts of each module on detection performance comprehensively, seven ablation experimental schemes were designed. For detection experiments, Scheme 1 replaced the corresponding module (RepNCSPELAN4) in the baseline network with the FasterCSPELAN4 feature extraction module, Scheme 2 added the SEAttention mechanism to the neck of the baseline network, Scheme 3 incorporated the Merge-NMS technique into the postprocessing part of the baseline network, Scheme 4 replaced the corresponding module in the baseline network with the FasterCSPELAN4 feature extraction module and added the SEAttention mechanism to its neck, Scheme 5 replaced the corresponding module in the baseline network with the FasterCSPELAN4 feature extraction module and added the Merge-NMS technique to its postprocessing part, Scheme 6 added the SEAttention mechanism to the neck of the baseline network and incorporated the Merge-NMS technique into its postprocessing, and Scheme 7 replaced the corresponding module in the baseline network with the FasterCSPELAN4 feature extraction module, added the SEAttention mechanism to its neck, and simultaneously incorporated the Merge-NMS technique into its postprocessing.

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TABLE X
DETECTION RESULT STATISTICS FOR ABLATION EXPERIMENTS OF THE SF_GELAN NETWORK

Scheme	Params	GFLOPS	mAP_50	mAP	mAP					
					Basketball	Track	Football	Tennis	Hockey	Baseball
Baseline	25.442	103.20	0.915	0.778	0.785	0.892	0.633	0.802	0.765	0.793
1	24.451	99.50	0.916	0.783	0.781	0.888	0.634	0.817	0.780	0.796
2	25.589	102.80	0.915	0.783	0.783	0.889	0.619	0.803	0.811	0.795
3	25.442	103.20	0.916	0.784	0.796	0.895	0.637	0.813	0.765	0.797
4	24.598	99.50	0.919	0.786	0.781	0.884	0.655	0.799	0.797	0.796
5	24.451	99.50	0.917	0.788	0.793	0.892	0.636	0.823	0.780	0.802
6	25.589	103.30	0.917	0.791	0.792	0.893	0.625	0.814	0.822	0.797
7	24.598	99.50	0.920	0.793	0.792	0.888	0.660	0.808	0.803	0.804

The specific detection results for each experimental scheme are presented in Table X. Compared with the baseline network, the addition of a single module could improve the detection accuracy of the network (measured by the mAP metric). The combination of any two modules yielded superior detection accuracy, and the simultaneous addition of all three modules achieved the best detection accuracy, which fully demonstrated the effectiveness and compatibility of the modules designed in this study. In terms of detection efficiency, the FasterCSPELAN4 and SEAttention modules were lightweighted. Therefore, compared with that of the baseline network, the detection efficiency changed little in all experimental schemes. The detection efficiency was slightly improved in all schemes that employed the FasterCSPELAN4 module.

From the perspective of single-module detection effects, adding the FasterCSPELAN4 module (Scheme 1) enhanced the network feature extraction capability and improved the detection accuracy for targets in various categories. Adding the SEAttention mechanism (Scheme 2) improved the ability of the network to balance categories and increased the detection accuracy for targets in categories with fewer instances (e.g., hockey fields). However, it might also lead to the loss of complex category information, thereby reducing the detection accuracy for targets in complex categories (e.g., football fields). The Merge-NMS module does not involve network structure. Adding the Merge-NMS technique to the postprocessing part (Scheme 3) could integrate high-precision detection results from feature maps of different scales effectively, improving the overall detection accuracy of the network.

A comprehensive comparison of the baseline, Scheme 1, Scheme 2, and Scheme 4 revealed that adding the SEAttention module to the baseline network (Scheme 3) led to a decrease in the detection accuracy for football fields. By contrast, adding the SEAttention module to Scheme 1 (Scheme 4) resulted in an increase in the detection accuracy for football fields. This was because the addition of the FasterCSPELAN4 module in Scheme 1 improved the aggregation of the channel feature information for complex categories. Subsequently, the addition of the SEAttention module assigned channel weights, which could highlight useful information and suppress background information, improving the detection accuracy for complex categories, such as football fields. In the baseline network, however, the channel feature extraction for complex categories was insufficient and dispersed. The addition of the SEAttention module led to some information loss and a decrease in the detection accuracy of complex categories. Overall, the SF_GELAN network with all three modules added (Scheme 7) achieved higher detection efficiency and accuracy than those of the baseline network with fewer parameters. The mAP_50 increased by 0.5%, and the mAP increased by 1.5%.

To present the roles of different modules in the network more intuitively, representative sample data were selected to create heatmaps for the baseline experiment, Scheme 1 (FasterCSPELAN4 feature extraction experiment), Scheme 2 (SEAttention attention mechanism experiment), and Scheme 7 (SF_GELAN network), as shown in Fig. 16.

The heatmap of Scheme 1 exhibits a more concentrated distribution of network attention than that of the baseline experiment, which is tightly focused on the primary feature regions of the target. This phenomenon indirectly reflects that the feature extraction capability of the network in Scheme 1 has been significantly enhanced, indicating that the introduction of the FasterCSPELAN4 module improves the feature extraction efficiency of the network and helps capture the important feature information of the target more accurately. Comparing the heatmap of Scheme 2 with that of the baseline experiment reveals that the former shows a more prominent display effect at the important information locations in the core of the target and that the relatively less important information at the edges of the target is slightly suppressed. This demonstrates that the application of the SEAttention module can optimize the ability of the network to screen and represent feature information, enabling it to focus more on the core features of the target and thereby improving the overall feature extraction effect.

Scheme 7 integrates the advantages of the FasterCSPELAN4 module and the SEAttention module, and its heatmap performance is superior. The heatmap clearly shows that the important areas of network attention are not only prominent but also highly concentrated on the primary features of various targets, which indicates that the network in Scheme 7 has superior performance in feature extraction and representation. It can focus on important feature information more accurately and achieve efficient extraction and precise representation of target features, demonstrating the performance improvement caused by the synergistic effect of the two modules. The visual conclusions obtained using the heatmaps are highly consistent with the results based on mAP accuracy

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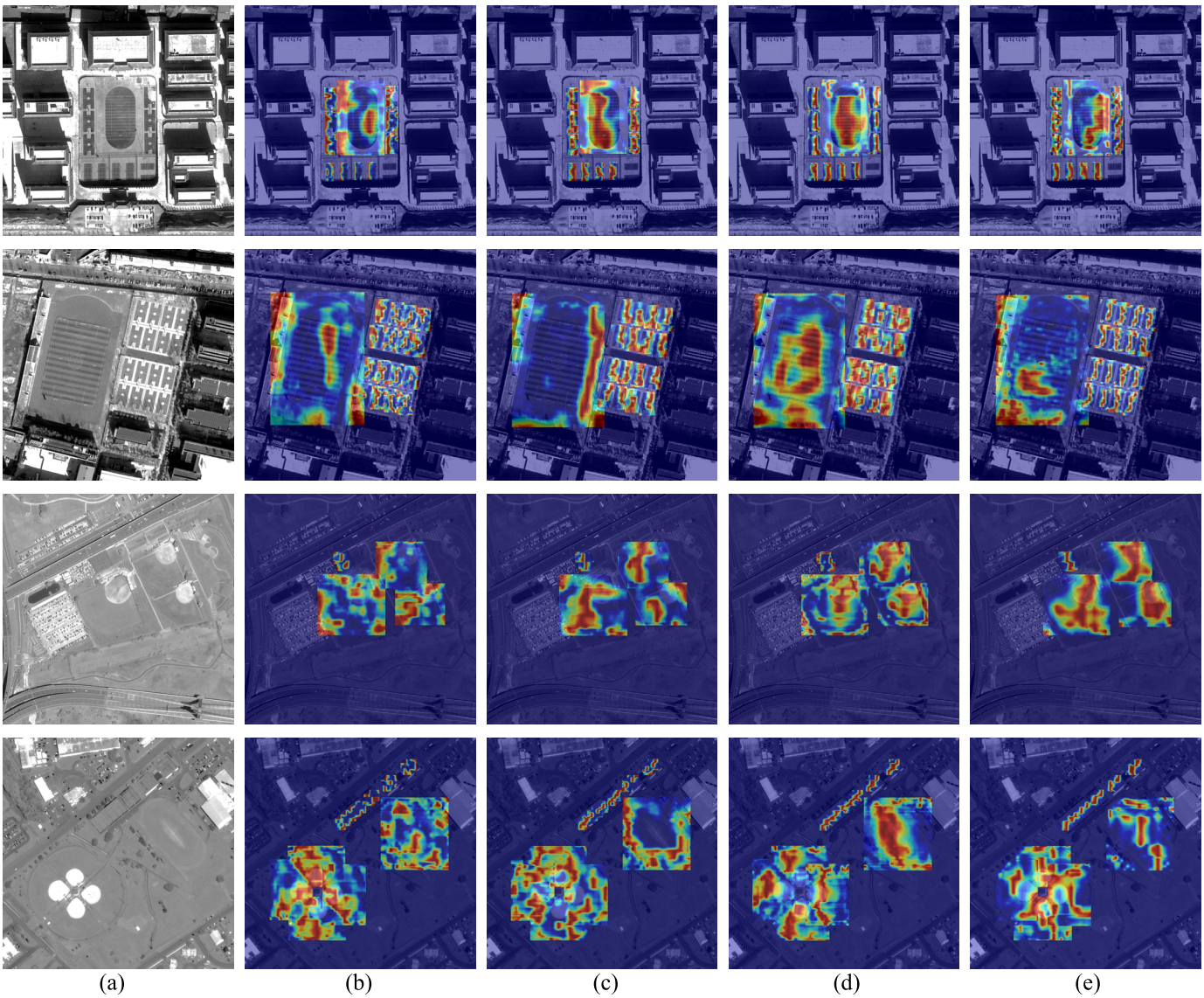


Fig. 16. Heatmaps of typical samples under different schemes. (a) Input image, (b) baseline, (c) Scheme 1, (d) Scheme 2, and (e) Scheme 7.

statistics, corroborating the effectiveness of the proposed algorithm. This further validates the advantages and reliability of the proposed method in network feature extraction and target recognition tasks, laying a solid foundation for subsequent in-depth research and practical applications.

3) Algorithm Comparison Experiments: To validate the performance of the SF_GELAN network further, three versions of the SF_GELAN network—large (SF_GELAN-B), medium (SF_GELAN-M), and small (SF_GELAN-S)—were generated using model scaling techniques. These versions were compared with corresponding versions of algorithms, such as YOLOv8, YOLOv9, YOLOv10, and YOLOv11 based on the SF_VHR_SSI dataset; the specific results are presented in Table XI. The SF_GELAN-B model has a width of 1.33 and a depth of 1.25, which corresponds to the largest versions (X and E) of the YOLO series. The SF_GELAN-M model has a width and depth of 1, corresponding to the L and C versions of the YOLO series. The SF_GELAN-S model has a width of 0.5 and a depth of 0.33, corresponding to the S version of the YOLO series.

For the small version, compared with the highest accuracy GELAN-S model in the state of the art (SOTA), SF_GELAN-S used fewer parameters, had higher detection efficiency, and achieved a 1.4% increase in mAP_50 and a 1.1% increase in mAP. Compared with the latest YOLOv11-S algorithm, SF_GELAN-S had lower detection efficiency but a larger increase in accuracy: a 5.1% increase in mAP_50 and a 6.7% increase in mAP. To display the performance of each model more intuitively, the detailed changes in precision, recall, mAP_50, and mAP during the training processes are illustrated in Fig. 17.

For the medium version, compared with the highest accuracy GELAN-C model in SOTA, SF_GELAN-M used fewer parameters, had higher detection efficiency, and achieved a 0.5% increase in mAP_50 and a 1.5% increase in mAP. Compared with the latest YOLOv11-L algorithm, SF_GELAN-M

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TABLE XI
DETECTION RESULT STATISTICS FOR DIFFERENT VERSIONS OF THE SF_GELAN NETWORK AND CORRESPONDING ALGORITHMS

Models	Params	GFLOPS	mAP_50	mAP	mAP					
					Basketball	Track	Football	Tennis	Hockey	Baseball
YOLOv8-S	11.138	28.70	0.849	0.691	0.737	0.858	0.492	0.746	0.587	0.726
GELAN-S	7.198	26.90	0.888	0.739	0.750	0.874	0.548	0.766	0.734	0.759
YOLOv9-S	9.747	39.60	0.887	0.733	0.742	0.871	0.574	0.763	0.678	0.767
YOLOv10-S	8.071	24.80	0.847	0.679	0.720	0.838	0.483	0.746	0.573	0.712
YOLOv11-S	9.430	21.60	0.851	0.683	0.724	0.856	0.487	0.744	0.570	0.718
SF_GELAN-S	**6.536**	**26.20**	**0.902**	**0.750**	**0.758**	**0.874**	**0.608**	**0.784**	**0.702**	**0.771**
YOLOv8-L	43.634	165.40	0.865	0.731	0.765	0.885	0.554	0.783	0.631	0.766
GELAN-C	25.442	102.80	0.915	0.778	0.785	0.892	0.633	0.802	0.765	0.793
YOLOv9-C	51.011	238.90	0.910	0.769	0.783	0.894	0.613	0.812	0.744	0.766
YOLOv10-L	25.775	127.20	0.867	0.730	0.750	0.867	0.575	0.789	0.635	0.764
YOLOv11-L	25.315	87.30	0.872	0.734	0.767	0.884	0.529	0.786	0.685	0.754
SF_GELAN-M	**24.598**	**99.50**	**0.920**	**0.793**	**0.792**	**0.888**	**0.660**	**0.808**	**0.803**	**0.804**
YOLOv8-X	68.158	258.10	0.878	0.739	0.770	0.884	0.580	0.790	0.643	0.765
GELAN-E	58.057	192.20	0.914	0.779	0.789	0.893	0.608	0.808	0.803	0.769
YOLOv9-E	69.416	244.90	0.914	0.781	0.793	0.894	0.623	0.810	0.792	0.776
YOLOv10-X	31.666	171.10	0.852	0.719	0.750	0.874	0.544	0.788	0.635	0.724
YOLOv11-X	56.881	195.50	0.890	0.756	0.771	0.894	0.589	0.808	0.706	0.766
SF_GELAN-B	**38.210**	**154.90**	**0.919**	**0.787**	**0.800**	**0.893**	**0.634**	**0.824**	**0.773**	**0.801**

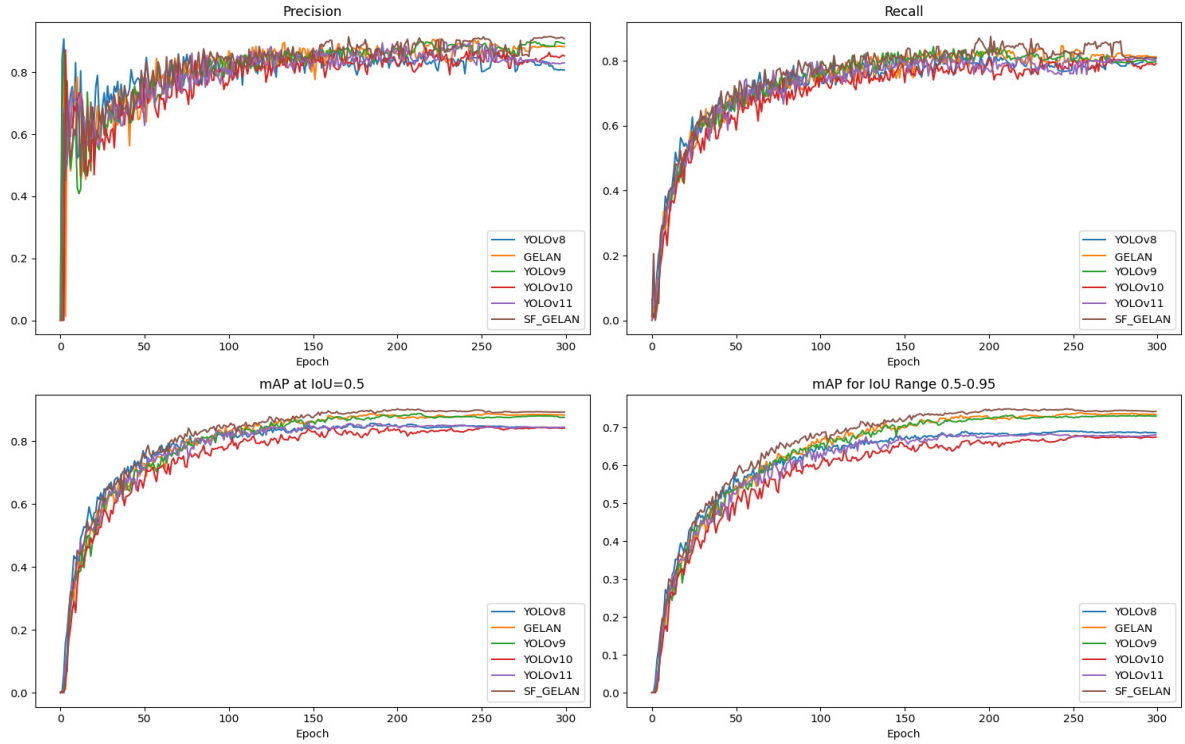


Fig. 17. Comparison of accuracy metric changes during training of small version for various algorithms.

had lower detection efficiency but a larger increase in accuracy: a 4.8% increase in mAP_50 and a 5.9% increase in mAP.

For the large version, compared with the highest accuracy YOLOv9-E model in SOTA, SF_GELAN-B used fewer parameters, had higher detection efficiency, and achieved a 0.5% increase in mAP_50 and a 0.6% increase in mAP. Compared with the latest YOLOv11-X algorithm, SF_GELAN-B showed significant improvements in both detection efficiency and accuracy. In terms of accuracy, a 2.9% and a 3.1% increase in mAP_50 and mAP was observed.

Overall, when the various versions of the SF_GELAN network are compared with their corresponding SOTA algorithms, SF_GELAN-S exhibited the most significant improvements (the advantages of lightweight design are more pronounced in smaller models) and SF_GELAN-M achieved the highest detection accuracy (owing to the limited number of samples

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TABLE XII
DETECTION RESULTS OF VARIOUS DETECTION ALGORITHMS ON THE DIOR DATASET

Models	mAP	mAP									
		C1	C2	C3	C4	C5	C6	C7	C8	C9	C10
RetinaNet (R-101) [15]	0.661	0.533	0.770	**0.693**	**0.850**	0.441	0.732	0.624	0.786	0.628	0.786
PANet (R-101) [15]	0.661	0.602	0.720	**0.706**	**0.805**	0.436	0.723	0.614	0.721	0.667	0.720
YOLOv8-S	0.656	0.743	0.685	**0.859**	**0.853**	0.376	0.841	0.521	0.787	0.659	0.652
GELAN-S	0.684	0.760	0.742	**0.871**	**0.875**	0.415	0.865	0.603	0.810	0.695	0.734
YOLOv9-S	0.685	0.760	0.764	**0.873**	**0.875**	0.409	0.868	0.593	0.812	0.700	0.727
YOLOv10-S	0.665	0.754	0.745	**0.871**	**0.859**	0.385	0.830	0.549	0.804	0.676	0.664
YOLOv11-S	0.672	0.748	0.736	**0.863**	**0.860**	0.399	0.827	0.559	0.803	0.671	0.688
SF_GELAN-S	**0.691**	**0.777**	**0.748**	**0.880**	**0.884**	**0.416**	**0.864**	**0.582**	**0.823**	**0.710**	**0.735**

Notes: R-101 denotes ResNet-101; RetinaNet and PANet are the two algorithms with the best detection performance in [15]. C1 represents the airplane category, C2 represents the airport category, C3 represents the baseball field category, C4 represents the basketball court category, C5 represents the bridge category, C6 represents the chimney category, C7 represents the dam category, C8 represents the expressway service area category, C9 represents the expressway toll station category, C10 represents the golf course category.

Models	mAP	mAP									
		C11	C12	C13	C14	C15	C16	C17	C18	C19	C20
RetinaNet (R-101) [15]	0.661	**0.766**	0.499	0.596	0.711	0.684	0.458	**0.813**	0.552	0.444	0.855
PANet (R-101) [15]	0.661	**0.734**	0.453	0.569	0.717	0.704	0.620	**0.809**	0.570	0.472	0.845
YOLOv8-S	0.656	**0.758**	0.529	0.524	0.605	0.832	0.596	**0.871**	0.440	0.439	0.545
GELAN-S	0.684	**0.788**	0.575	0.550	0.603	0.858	0.593	**0.878**	0.464	0.445	0.547
YOLOv9-S	0.685	**0.790**	0.567	0.558	0.602	0.858	0.593	**0.876**	0.490	0.446	0.545
YOLOv10-S	0.665	**0.762**	0.543	0.523	0.611	0.855	0.607	**0.879**	0.400	0.452	0.550
YOLOv11-S	0.672	**0.774**	0.548	0.538	0.606	0.853	0.600	**0.878**	0.483	0.450	0.550
SF_GELAN-S	**0.691**	**0.787**	**0.575**	**0.549**	**0.616**	**0.861**	**0.601**	**0.889**	**0.488**	**0.462**	**0.564**

Notes: R-101 denotes ResNet-101; RetinaNet and PANet are the two algorithms with the best detection performance in [15]. C11 represents the track field category, C12 represents the harbor category, C13 represents the overpass category, C14 represents the ship category, C15 represents the stadium category, C16 represents the storage tank category, C17 represents the tennis court category, C18 represents the train station category, C19 represents the vehicle category, and C20 represents the windmill category.

in the SF_VHR_SSI dataset, deeper and larger models do not necessarily yield better results). These findings are closely related to the lightweight design of the FasterCSPELAN4 and SEAttention modules, as well as the size and characteristics of the SF_VHR_SSI dataset. These results validate the rationality and effectiveness of the SF_GELAN network design.

4) Algorithm Generalization Experiments: Although this study focused on the task of detecting sports fields, the constructed SF_VHR_SSI dataset, as a multitype and fine-grained remote-sensing image target detection dataset, possesses the general characteristics of remote-sensing image datasets, including being multiscale, being multiangle, and having a complex background. Given that the SF_GELAN network performs well on this dataset, it theoretically has the potential to perform well on other remote-sensing datasets. At the algorithm level, the SF_GELAN network is developed based on the YOLO framework, which is both general and applicable. The FasterCSPELAN4 module and the SEAttention module are introduced to optimize the feature extraction and representation efficiency of the algorithm, enhancing the robustness of the algorithm as well as the detection accuracy and efficiency. The general architecture of the algorithm is completely retained, which implies that the SF_GELAN algorithm theoretically has good generalization ability.

To verify the generalization ability of the research results further, the classic DIOR dataset [15] was selected for general object detection experiments based on SF_GELAN-S, and the detection results were compared with YOLOv8-S, GELAN-S, YOLOv9-S, YOLOv10-S, YOLOv11-S, and the best results in [15] and analyzed. The detailed data are shown in Table XII. The DIOR dataset contains 20 categories, including airports, basketball courts, bridges, ports, dams, and ships, with a total of 23 463 samples. For the present study, the dataset was divided into a training set (14 077 samples), validation set (4694 samples), and test set (4692 samples) in a 6:2:2

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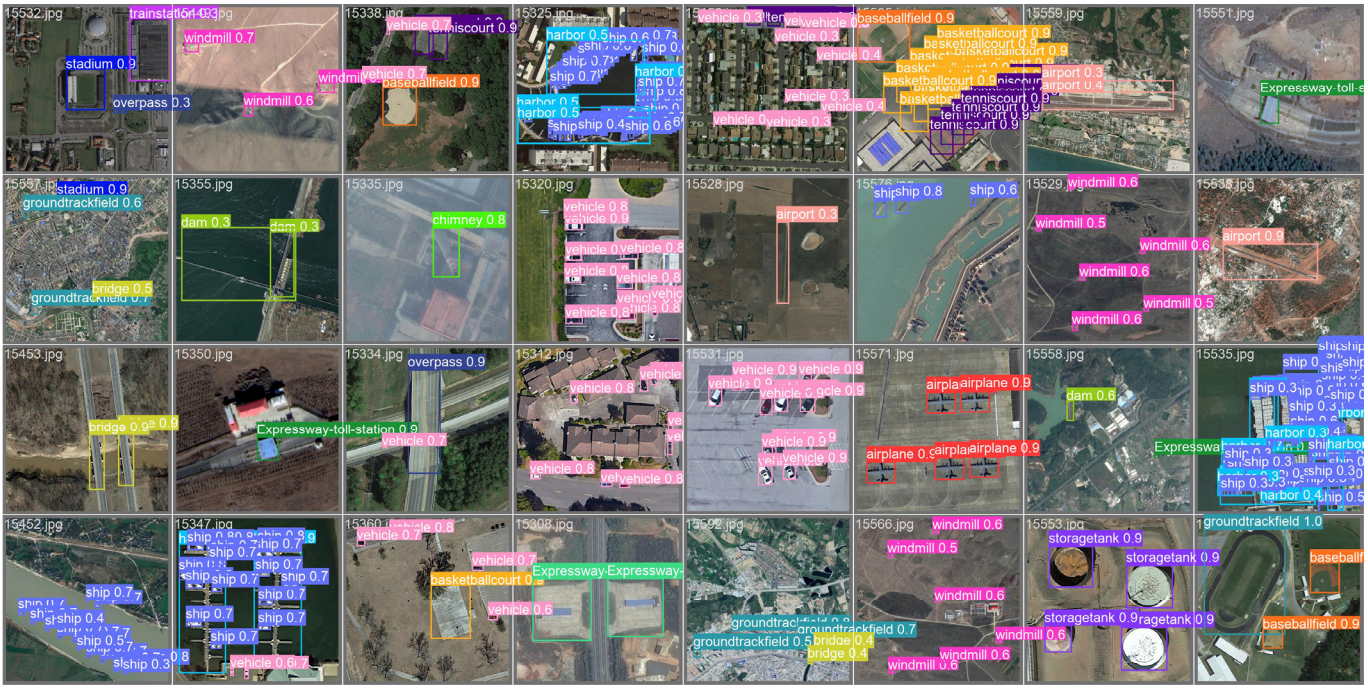


Fig. 18. Visualization of typical sample detection results of the SF_GELAN-S algorithm on the DIOR dataset.

ratio. The experimental software and hardware conditions are shown in Table II. No pretrained weight files were used in the experiments. The input image size was uniformly scaled to 640 × 640, and the training batch size was uniformly set to eight. All other hyperparameters were set to the default values of the platform algorithm.

Compared with RetinaNet and PANet, which had the best detection performance in [15], SF_GELAN-S has more concise parameters and significantly better detection efficiency. Moreover, the overall mAP is 3.0% higher, indicating that SF_GELAN-S is comparable to classic algorithms, such as RetinaNet and PANet, and has the characteristics of general detection and good data applicability. In terms of single-category mAP, the accuracy for sports field categories (such as baseball field, basketball court, track field, and tennis court) improved significantly. The accuracy for categories, such as airplanes, chimneys, highway service areas, and harbors, also increased significantly, highlighting the strong category applicability of the SF_GELAN-S algorithm. However, the accuracy for categories, such as bridges, train stations, and ships, declined to some extent, reflecting that the detection performance of SF_GELAN-S for slender targets must be enhanced further.

Compared with the YOLO series of algorithms, SF_GELAN-S ranks first in mAP, once again confirming the advantages and reliability of the proposed algorithm in the task of target recognition and its applicability in the target field. Compared with GELAN-S, which has the highest accuracy in SOTA, the overall mAP of SF_GELAN-S is increased by 0.7%, which is lower than the 1.1% improvement on the SF_VHR_SSI dataset. This phenomenon primarily results from the limitations of slender targets in the DIOR dataset. To make the detection results more intuitive, some typical samples were selected to visually present the detection results of the SF_GELAN-S algorithm, as shown in Fig. 18.

V. Conclusion and Prospects

To address the issue of high-precision extraction of sports fields from very-high-resolution satellite imagery, the first multicategory, fine-grained sports field dataset, SF_VHR_SSI, was constructed based on real submeter stereo satellite imagery. The richness and diversity of the dataset were analyzed based on information, such as data sources, coverage, and acquisition times. Extensive benchmark experiments were conducted using more than 50 models of over a dozen classic algorithms, demonstrating the scientific and challenging nature of the dataset. Building on this foundation and considering the characteristics of the dataset, a lightweight feature extraction module (FasterCSPELAN4) and an attention module (SEAttention) were developed. Additionally, a scientific Merge-NMS postprocessing technique was designed, and the SF_GELAN network, which combines high efficiency and high precision, was constructed. Using model scaling techniques, large, medium, and small versions of the network were developed for different applications. Comparisons showed that all three versions of the SF_GELAN network achieved SOTA performance on SF_VHR_SSI.

Future work will focus on enriching and expanding the SF_VHR_SSI dataset. Additionally, the number of samples and instances will be increased to support the training of larger models. Moreover, new categories will be established according to the needs of remote-sensing mapping to expand the application scope of the dataset. The next step will involve exploring multiscale convolution mechanisms in combination with lightweight techniques to enhance the context extraction capabilities of the network and further improve model detection accuracy.

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Dashuai Shang received the M.S. degree in science of photogrammetry and remote sensing from the Institute of Surveying and Mapping, Zhengzhou, China, in 2012.

His research interests include digital photogrammetry and laser point cloud processing.



Yongsheng Zhang received the Ph.D. degree in science of surveying and mapping from the Institute of Surveying and Mapping, Zhengzhou, China, in 1990.

He has developed several theories in the application of aerospace remote sensing information processing. He has authored or co-authored over 200 papers. His research interests include 3-D positioning of satellite remote sensing images, space-based multisource remote sensing information integration, and precision calibration of on-obrit satellite.

Prof. Zhang is an Editorial Board Member of Acta Geodaetica et Cartographica Sinica.



Chenguang Dai received the Ph.D. degree in science of photogrammetry and remote sensing from the Institute of Surveying and Mapping, Zhengzhou, China, in 2008.

His research interests include digital photogrammetry and laser point cloud processing.



Lei Zhang received the M.E. degree in science of surveying and mapping from the Institute of Surveying and Mapping, Zhengzhou, China, in 2021.

His research interests include intelligent interpretation of remote sensing images and geospatial intelligence.

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